
Principled Thoughts for Latent Recursive LLM Systems

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 fard-lab.github.io/REST

Preprint

Abstract. Large language models can reason in continuous space instead of decoded text, by recurring on their own hidden states or by passing those states between agents, while training supervises only the Cross-Entropy (CE) of the final decoded answer and does not constrain the thought. Theoretical and empirical analyses establish and confirm four failures of CE-only training that lead to a lower probability of the correct answer such as collapsing thoughts across distinct questions and retaining irrelevant information. We introduce **REST** (REpresentation-Supervised Thoughts), a training objective that turns four properties of a valid thought representation (causality, minimality, separability, and stability) into differentiable losses added to CE. We instantiate it in latent single-agent and multi-agent systems, without architectural changes or added parameters at inference. Across 7 benchmarks spanning mathematics, science, medicine, and code generation, with the same training data, compute, and latent budget, REST increases accuracy over CE-only training across agent settings and model sizes by up to 7.5 percentage points and convergence on a final answer by 30%. Furthermore, REST thoughts encode more of what is required to achieve the correct answer, and decoding them better recovers the intended output of the agent, which makes latent communication easier to interpret.

1 Introduction

Chain-of-Thought (CoT) prompting enables Large Language Models (LLMs) to solve complex problems by generating each intermediate reasoning step as explicit text [45]. However, expressing every step through tokens of a discrete vocabulary confines reasoning to paths that can be written in words [26]. The resulting reasoning traces are also long, which raises inference cost and leads models to allocate excessive computation to simple problems [5].

To resolve these limitations, a growing line of work reasons directly in the continuous latent space of LLMs rather than through decoded text [2, 19, 40, 41, 46, 53]. Within a single model, this is realized by training the model to consume its own hidden states as the next reasoning step instead of a token embedding [19, 27, 40]. From one model to another, hidden states or KV caches have instead been used for inter-agent communication [15, 31, 54]. Recent systems unify both, with multiple agents reasoning and communicating in latent space [59] and repeating this collaboration over several rounds of recursion [58]. The main objective is usually Cross-Entropy (CE) of the final decoded answer. As these systems scale in agents and rounds, each later agent builds on the thoughts it receives, and error compounds through the remainder of the system. Existing representation-level supervision targets the text the thought replaces, which CODI does by matching the hidden states that text induces [40] or SIM-COT by decoding that text back from the thought [46]. However, supervising only on the target text does not dictate the relation between two thoughts nor the distribution of outputs the model can generate. Both requirements are among the four properties that define a valid

thought representation [39].

A valid thought representation should preserve what is needed to reproduce the output (Causality), discard what is irrelevant to that output (Minimality), remain distinguishable across semantically distinct inputs (Separability), and reflect the distribution of possible outputs rather than one sample from it (Stability). An audit of existing latent representations finds that every candidate violates at least one property. The audit evaluates existing methods, and does not perform training to target these properties. Training with CE only does not enforce them either, since a model receives no signal as long as the correct answer maintains high probability (Section 2.1).

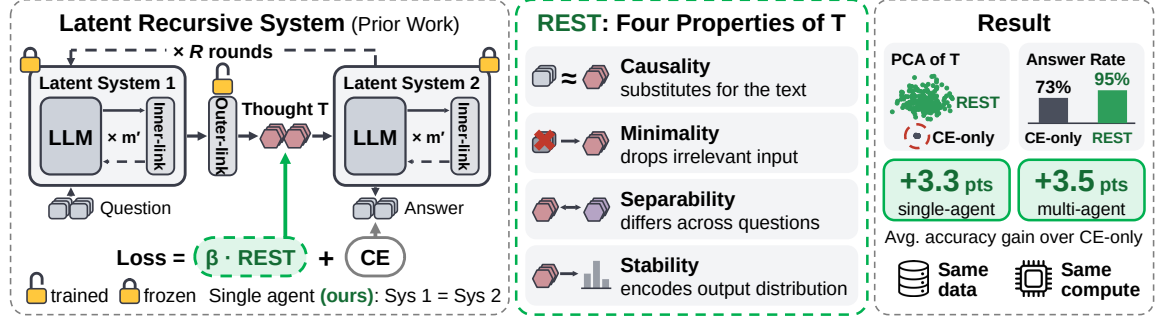


Figure 1. Overview of REST. Latent systems pass a thought T through a trained outer link (left). REST adds a β -weighted loss built from the four properties of T to the CE objective (middle). CE-only thoughts collapse and REST improves accuracy under the same data and compute (right).

We introduce **REpresentation-Supervised Thought(s) (REST)**. REST translates each of these four properties into a differentiable loss term (Appendix A.1) and adds it to the total loss alongside CE. Each term is weighted by a hyperparameter β , added on top of the existing CE loss. Training then targets the thought representation as well as the decoded answer (Figure 1). REST is studied on a latent recursive system in which the base LLMs are frozen [58]. We extend this construction from the multi-agent setting in which it was proposed to a single-agent setting, where a model recurs on its own hidden states. In addition, no changes to architecture or inference procedure are required, allowing it to be added to existing latent recursive systems without modification.

REST increases downstream accuracy across MATH500 [29], GPQA-Diamond [36], MedQA [23], AIME2025/2026 [12, 52], LiveCodeBench-v6 [22], and MBPP+ [30], spanning mathematical, scientific, and code-generation domains. Averaged over all REST configurations, it improves accuracy over CE only by +3.5 points in the multi-agent setting and +3.3 in the single-agent setting, under a matched training budget, the same training data, and the same latent budget. The best configurations reach +7.5 and +6.5 points, respectively. Through our analysis, we confirm that the trained thoughts exhibit the properties that were targeted, which in turn lower-bound the probability of the correct answer (Appendix A.2). REST decodes 15.4% more tokens on average at inference, which we attribute to a higher rate of convergence on a final answer (Section 4.4).

Our contributions can be summarized as follows:

- We identify four failures of CE-only thoughts that lead to a lower probability of the correct answer (Section 2.1). In trained systems, CE-only thoughts collapse across distinct questions and retain irrelevant information (Section 4.4).
- We introduce REST, which translates four theoretically motivated properties of thought representations [39] into differentiable loss terms added to the CE objective, and we extend latent recursive systems [58] from the multi-agent setting to a single agent that recurs on its own hidden states.
- REST improves accuracy over the CE baseline across mathematical, scientific, medical, and code-generation benchmarks in both the single-agent and multi-agent settings, under matched training and latent budgets, and its gains remain robust across loss weights.

2 Background

Setup and Notation. Sets and operators are denoted with $(\mathcal{V}, \mathcal{R}_{\text{in}}, \mathcal{R}_{\psi})$, random variables, matrices, and scalar-valued functions in upper case (Y, H, J) , and scalars, indices, token sequences, and single vectors in lower case (m, t, u, e_i) , with bold for the thought $\mathbf{T}$ and vectors derived from it $(\mathbf{t})$. A next-token distribution is lower case (p, q) while the sequence is the matching upper case (P, Q) .

We adopt recursive multi-agent systems [58], and extend their approach to the single-agent setting to study REST. Let $f_{\theta}(\cdot)$ denote an auto-regressive transformer model with vocabulary $\mathcal{V}$ and hidden size d_h . Given a token sequence u with input embeddings $E(u) = [e_1, \dots, e_m] \in \mathbb{R}^{m \times d_h}$, a forward pass yields last-layer hidden states $H = [h_1, \dots, h_m] \in \mathbb{R}^{m \times d_h}$ and, at each position i , the next-token distribution $\text{softmax}(h_i W_{\text{out}})$ over $\mathcal{V}$. A multi-agent system $\mathcal{S}$ comprises N agents $\mathcal{A} = \{A_1, \dots, A_N\}$, where each A_i corresponds to f_{θ_i} , where every θ_i is frozen.

Hidden states re-enter embedding space through two links [58]. An inner link $\mathcal{R}_{\text{in}}$ maps an agent’s hidden state back to its own input space, which lets A_i take a latent step instead of decoding a token. It is trained per agent by aligning $\mathcal{R}_{\text{in}}(H)$ onto the embeddings of the ground-truth text. An outer link $\mathcal{R}_{\psi}$ maps the output of $\mathcal{R}_{\text{in}}$ into the input space of another agent. $\mathcal{R}_{\text{in}}$ ’s objective only aligns hidden states to the input embedding space, whereas all outer links are trained through CE on the target answer text at the final agent.

A latent thought of a producer $A_i \in \mathcal{A}$ to a consumer A_j composes both links as $\mathbf{T} = \mathcal{R}_{\psi}(\mathcal{R}_{\text{in}}(H_u)) \in \mathbb{R}^{m' \times d_h}$, where $H_u \in \mathbb{R}^{m \times d_h}$ denotes the producer’s last-layer hidden states at the m positions of its own output $u \in \mathcal{V}^m$, and m' is a latent steps budget for transferred thoughts.

Definition 2.1 (Transfer). For a consumer $A_j \in \mathcal{A}$, let E_{pre} and E_{post} denote the embedding blocks of a prompt that surround the transferred content, both independent of ψ . For every block Z of d_h -dimensional embedding vectors, the consumer performs a forward pass on $[E_{\text{pre}}; Z; E_{\text{post}}; E(v)]$ where v is the teacher-forced target answer for the consumer. The transfer can be through latent thoughts for $Z = \mathbf{T}$ and through textual embeddings for $Z = E(u)$.

Training and inference differ in how H_u is obtained. During training, u is the producer’s ground-truth output, and one teacher-forced pass over the producer’s prompt concatenated with u yields H_u at u ’s positions. During inference no u is available in advance, so the producer starts from its prompt alone and takes m' latent steps. At each step, $\mathcal{R}_{\text{in}}$ maps the last position’s hidden state into the next input embedding, and $H_u = [h_1, \dots, h_{m'}]$ is the sequence of hidden states over these steps.

Roles and Hops. The system chains a planner A_1 , a refiner A_2 and a solver A_3 (Figure 1, Table 1). The planner proposes an approach to the question, the refiner revises that approach, and the solver produces the answer. Each agent takes m' latent steps through $\mathcal{R}_{\text{in}}$, and $\mathcal{R}_{\psi}$ maps the resulting hidden states into the next agent’s input space. Every handoff of this kind is one transfer of Definition 2.1.

Recursion. Once the solver finishes, its thought returns to the planner and the chain repeats for a further round [58]. Let R denote the number of rounds, let $\mathbf{T}^{(r,k)}$ denote the thought that A_k receives at round r , and let $\mathcal{H}$ collect the pairs (r, k) at which a transfer occurs. The planner of the first round is the one agent that conditions on the question without a transferred thought. Every later agent conditions on both its own input context and the thought it receives. The solver of the final round decodes v as text, and every earlier hop stays in latent space.

2.1 Limitations of Answer-Level Supervision

Substitution Divergence. Let $P(\cdot | Z)$ denote the distribution the consumer assigns to complete targets under a transferred block Z . A consumer answers under either block of Definition 2.1. Let CE_{lat} and CE_{txt} denote the per-token CE it incurs on a target v under each one. Let $\Lambda(v) = \text{CE}_{\text{lat}}(v) - \text{CE}_{\text{txt}}(v)$ denote the cost of substituting the thought for the producer’s text. We can represent this as a divergence between the two transfers over the targets the producer’s text induces,

$$\mathbb{E}_{v \sim P(\cdot | E(u))} [\Lambda(v)] = \frac{1}{n} D_{\text{KL}}(P(\cdot | E(u)) \| P(\cdot | \mathbf{T})), \quad (1)$$

where n is the length of v (Appendix A.2.1). This divergence can be zero only when both transfers lead the consumer agent to the same distribution. The probability the consumer assigns to the correct target is reduced by a factor of $\exp(-n\Lambda(v))$. A longer target is therefore penalized more for the same divergence per token. This effect compounds for each hop $A_i \rightarrow A_j$ in the pipeline. Empirically, CE-only does not reach the accuracy of the oracle text from the producer agent (Figure 5(b)).

Information Loss. When a consumer agent solves a question, it may solve a large part on its own (Table 6), and on those questions CE would be close to its minimum. Gradients updating the thought for those examples would be small compared to others. Therefore, the output from a producer agent may not be preserved as ψ may not learn from such examples. The consumer agent is then left with less information than when given the producer’s generated output (Proposition A.13). In addition, since the transfer is further limited to a budget of m' positions, ψ may learn to encode irrelevant information due to such examples since their supervision signal is too weak. Empirically, CE-only encodes more content from the producer agent’s input than its output (Figure 5(a)).

Thought Collision. If training resulted in two different examples having colliding thoughts, then a consumer agent must answer both with nearly the same distribution even if the two target texts are different. This imposes a lower bound on their average CE (Proposition A.17). Under total collapse, where the two thoughts are identical, the optimal solution would be assigning equal probability to both targets, which would mean it is random chance. CE evaluates every example on its own target. Summing over examples does not change this, since each term depends on one thought only. A collision raises the loss on both examples, and CE cannot distinguish this from two difficult examples. Empirically, CE-only collapses distinct questions together (Figure 6).

Producer Uncertainty. Generated text from a producer used as supervision targets would not encode the level of certainty for that output. At each step, it may be choosing between alternative solutions of equal probability. CE supervises only the consumer’s final answer, and it does not require the thought to encode this certainty. The loss would be higher for a thought that doesn’t encode the possible output distribution from a producer agent (Proposition A.19).

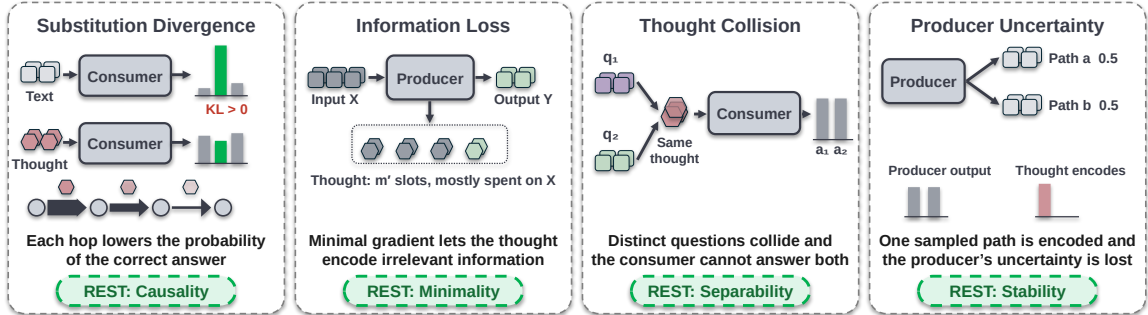


Figure 2. Why CE is not enough. Four failures of CE thoughts and the property that addresses each.

3 REST: Representation-Supervised Thought(s)

Every agent base LLM f_{θ_i} stays frozen alongside $\mathcal{R}_{in}$. The outer link $\mathcal{R}_{\psi}$ is the component that is trained with CE, thus we keep $\mathcal{R}_{in}$ frozen and only train $\mathcal{R}_{\psi}$ to isolate the effect of REST. CE is kept as part of the total loss function to maintain legible text outputs when answering. For each term, we start with the definition of the property and arrive at the loss function below (proofs in Appendix A.1).

3.1 Principled Thoughts

Let n denote the length of the target v of Definition 2.1, let $v_{<t} = (v_1, \dots, v_{t-1})$ denote its prefix at position $t \in \{1, \dots, n\}$, and let $p(\cdot | v_{<t}, Z)$ denote the consumer agent’s next-token distribution at that position under the transferred block Z .

Causality requires that **T** hold the required information about the text generated by a producer agent when given to a consumer agent without altering what a consumer agent would have predicted.

We enforce this property by penalizing the divergence between the two transfers of Definition 2.1,

$$\mathcal{L}_{\text{caus}}(\psi) = \frac{1}{n} \sum_{t=1}^n D_{\text{KL}}\left(p(\cdot \mid v_{<t}, E(u)) \parallel p(\cdot \mid v_{<t}, \mathbf{T})\right). \quad (2)$$

Minimality requires that $\mathbf{T}$ should remove irrelevant information that was present in the input of the producer agent while maintaining relevant information relative to its output. Let X and Y denote the producer’s input and output as random variables, the latter realized by the sequence u (Definition 2.1), and let $\text{CE}(Y \mid \mathbf{T})$ and $\text{CE}(X \mid Y, \mathbf{T})$ denote the CE terms calculated for the consumer agent,

$$\mathcal{L}_{\text{min}}(\psi) = \lambda_1 \text{CE}(Y \mid \mathbf{T}) - \lambda_2 \text{CE}(X \mid Y, \mathbf{T}), \quad (3)$$

where $\lambda_1, \lambda_2 > 0$ are scalars. Since a thought representation $\mathbf{T}$ from the producer agent may encode information from its input and output, this loss term would penalize a representation that would encode irrelevant information from its input relative to the information present in its output.

Separability requires that two thought representations for semantically distinct outputs should be distinguishable or separable to represent that they encode semantically distinct information. In order to apply this on $\mathbf{T}$, which can be a sequence of vectors, we utilize an attention pool to produce one vector for each sequence. Let $\mathbf{t}$ and $\mathbf{t}_1, \dots, \mathbf{t}_K$ in $\mathbb{R}^{d_h}$ denote the thoughts pooled over the m' positions of the current example and of the K training examples that precede it, respectively, and let $\text{sim}(a, b)$ denote the cosine similarity. We penalize the similarity of $\mathbf{t}$ to each $\mathbf{t}_k$,

$$\mathcal{L}_{\text{sep}}(\psi, \omega) = \log \sum_{k=1}^K \exp\left(\text{sim}(\mathbf{t}, \mathbf{t}_k) / \tau\right), \quad (4)$$

where $\tau > 0$ is a scalar for temperature and ω denotes the parameters of the attention pooling.

Stability requires encoding the output distribution rather than one sequence. Sampling that distribution would require many outputs at every stage of the system. Estimating the entropy on the other hand can track the property without bias (Appendix A.1.4). Let $q(\cdot \mid u_{<s})$ denote the producer’s next-token distribution at position $s \in \{1, \dots, m\}$ of its own output u , let $\mathbb{H}$ denote Shannon entropy, and let

$$\hat{\mathbb{H}}(u) = \frac{1}{m} \sum_{s=1}^m \mathbb{H}(q(\cdot \mid u_{<s})) \quad (5)$$

denote the producer’s mean predictive entropy along u . We enforce the property through the squared error between eq. (5) and an estimate of its value through a learned probe that receives $\mathbf{T}$,

$$\mathcal{L}_{\text{stab}}(\psi, \omega) = \left(g_{\omega}(\mathbf{T}) - \hat{\mathbb{H}}(u)\right)^2, \quad (6)$$

where g_{ω} is an attention pooling over the m' positions of $\mathbf{T}$ with an affine map from $\mathbb{R}^{d_h}$ to $\mathbb{R}$.

3.2 Multi-Agent Systems

REST adds one term at every transfer of Definition 2.1 and leaves the CE of the system unchanged,

$$\mathcal{L}(\psi, \omega) = \mathcal{L}_{\text{CE}}(v \mid \mathbf{T}^{(R,N)}) + \beta \frac{1}{|\mathcal{H}|} \sum_{(r,k) \in \mathcal{H}} \mathcal{L}_{\text{prop}}(\psi, \omega; \mathbf{T}^{(r,k)}), \quad (7)$$

where $\mathcal{L}_{\text{prop}} \in \{\mathcal{L}_{\text{caus}}, \mathcal{L}_{\text{min}}, \mathcal{L}_{\text{sep}}, \mathcal{L}_{\text{stab}}\}$, and $\mathbf{T}^{(R,N)}$ is the thought that the solver of the final round receives. ω is present only in $\mathcal{L}_{\text{sep}}$ and $\mathcal{L}_{\text{stab}}$. The property term is averaged over each transfer, therefore receiving a gradient from its own term in addition to the final CE.

3.3 Single-Agent Systems

It is important to note that Definition 2.1 permits $A_i = A_j$, which corresponds to our extension to the single-agent setting. REST extends to a single agent reasoning in latent space, rather than being restricted to a pipeline of several agents.

Self-Loop. A single solver acts as both the producer and the consumer, with no planner and no refiner (Figure 3). The solver first runs on its own input context and computes last-layer hidden states. These states pass through the frozen inner link $\mathcal{R}_{in}$ and the trained outer link $\mathcal{R}_{\psi}$, which gives the thought $\mathbf{T}$ of Section 2. The solver then runs a second time, conditioned on both its own input context and $\mathbf{T}$. Each further round repeats the second pass, and the hidden states it computes form the next round’s thought. Equation (7) applies at $N = 1$. The two links act at different points of the system. $\mathcal{R}_{in}$ runs at every one of the m' latent steps and lets the solver take a latent step instead of decoding a token, whereas $\mathcal{R}_{\psi}$ runs once per round. Removing $\mathcal{R}_{in}$ therefore removes latent reasoning itself rather than one trainable component. Keeping it frozen leaves ψ as the only parameter that varies between CE and REST, and between the single- and multi-agent settings.

4 Evaluation

4.1 Experimental Setup

Systems and Baselines. We evaluate REST on open-weight LLMs across the Qwen [34, 35, 48, 49], Llama [18], and Gemma [17] model families for heterogeneous agent collaborations. Table 1 lists the model assigned to each role. For baseline comparisons, we evaluate CODI [40] and SIM-CoT [46] adapted as loss terms added to CE, and the frozen LLMs as a text baseline (Table 6). CE only is the unmodified latent recursive system that REST builds on, and Best pair combines causality and minimality. Detailed implementations and hyperparameters are in Appendix B.2 and Appendix B.

Data. We adopt Sequential-Math [58] for training, constructed by rewriting question-answer pairs curated from s1K [33], m1K [20] into role-specific texts. Evaluation covers four domains, mathematics with MATH500 [29], AIME2025 [52], and AIME2026 [12], science with GPQA-Diamond [36], medicine with MedQA [23], and code generation with MBPP+ [30] and LiveCodeBench-v6 [22]. Additional details in Appendix B.1.

Training and Inference. For training, we freeze all agent parameters and update only ψ and ω . The training objective adds a weighted term to the loss function. AdamW is used under a cosine learning rate schedule. During inference, we follow each model’s official recommended sampling settings, and a lower temperature for code generation than for other reasoning tasks. We average values across three training seeds. The Avg. Change column reports the average change against the CE-only baseline across the benchmarks of one system, and each system is reported at the recursion round that performs best for it, with the baseline taken at that same round for fair comparison.

4.2 Single-Agent Evaluation

Table 2 shows every property term improving accuracy over the CE-only objective in this setting. The effect can be attributed to the thought rather than collaboration among agents. A practitioner is also more likely to already be running a single model system when considering costs. In the Light system, the token count is lower than CE, while it is higher in the multi-agent setting. Among the four properties, minimality delivers the largest gain in the single-agent setting. In the Light system, minimality exceeds the baseline on every task while decoding fewer tokens overall, indicating that its accuracy gain does not come from longer generation. This is consistent with the structure of

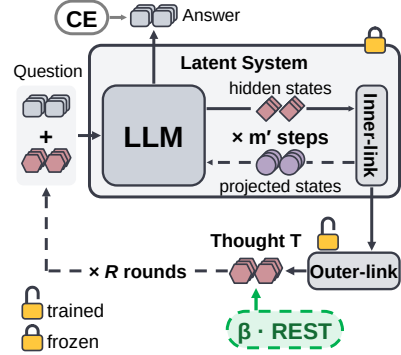


Figure 3. Single agent. $\mathcal{R}_{in}$ runs at each of the m' steps, and $\mathcal{R}_{\psi}$ once per round.

Table 1. Light and Scaled systems.

System	Role	Model
Light	Planner	Qwen3-1.7B
	Refiner	Llama-3.2-1B-Instruct
	Solver	Qwen2.5-Math-1.5B-Instruct
Scaled	Planner	Gemma-3-4B-it
	Refiner	Llama-3.2-3B-Instruct
	Solver	Qwen3.5-4B

Table 2. Single-agent, Light vs Scaled, at $r = 1$ and $r = 3$ respectively, each row at its best β .

Method	Metric	Math500		AIME2025		AIME2026		GPQA-D		MedQA		Code Gen.		Avg. Change	
		Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled
CE only	Acc.	70.6	80.9	27.8	65.6	12.2	72.2	27.3	63.3	27.1	79.2	27.7	35.5	Base	Base
	Token	557	862	905	7226	1016	6967	911	1709	1177	739	477	1289	Base	Base
REST (ours), SINGLE PROPERTY															
Causality	Acc.	72.4	80.4	23.3	78.9	16.7	83.3	23.9	63.5	29.7	79.4	32.9	40.2	$\uparrow 1.0$	$\uparrow 4.8$
	Token	551	1008	907	8677	1008	8165	851	2216	1113	955	570	1662	-0.9%	$+20.7\%$
Minimality	Acc.	72.1	81.1	30.0	78.9	20.0	86.7	27.6	63.8	29.7	83.0	31.5	42.3	$\uparrow 3.0$	$\uparrow 6.5$
	Token	546	1053	906	10256	936	10390	872	2526	1075	1162	561	1850	-2.9%	$+44.9\%$
Separability	Acc.	73.1	80.2	26.7	76.7	17.8	83.3	28.8	61.1	27.7	81.7	30.2	39.4	$\uparrow 1.9$	$\uparrow 4.3$
	Token	551	1056	918	8657	987	8033	901	2556	1083	1087	488	1764	-2.3%	$+23.2\%$
Stability	Acc.	72.1	79.1	27.8	83.3	15.6	88.3	28.1	64.4	29.8	79.8	29.1	40.4	$\uparrow 1.6$	$\uparrow 6.4$
	Token	538	1049	921	9626	944	9174	845	2464	1051	1071	645	1783	-2.0%	$+33.9\%$
REST (ours), COMPOSITION OF PROPERTIES															
Best pair	Acc.	69.7	79.4	23.3	80.0	16.7	86.7	26.1	57.6	29.6	79.0	33.5	38.6	$\uparrow 1.0$	$\uparrow 4.1$
	Token	542	1065	876	10503	1039	10728	826	2605	1083	1179	553	1632	-2.4%	$+47.5\%$
All properties	Acc.	72.4	78.4	33.3	73.3	16.7	76.7	23.7	60.6	31.0	79.3	32.8	37.0	$\uparrow 2.9$	$\uparrow 1.4$
	Token	569	981	934	11053	984	10338	953	1744	1130	1001	555	1375	$+1.6\%$	$+41.0\%$

the self-loop, in which the producer’s input contains the same question and instructions that the consumer already conditions on when receiving $\mathbf{T}$. As a result, any input content encoded in $\mathbf{T}$ is redundant and occupies latent positions that would otherwise carry the producer’s output.

4.3 Multi-Agent Evaluation

Table 3. Multi-agent, Light vs Scaled, at $r = 1$ and $r = 3$ respectively, each row at its best β .

Method	Metric	Math500		AIME2025		AIME2026		GPQA-D		MedQA		Code Gen.		Avg. Change	
		Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled
CE only	Acc.	71.1	86.6	22.2	80.0	17.8	60.0	24.9	59.1	28.9	79.7	32.5	33.5	Base	Base
	Token	550	1252	849	10485	884	5420	749	2572	868	829	604	1029	Base	Base
REST (ours), SINGLE PROPERTY															
Causality	Acc.	77.8	86.8	27.8	80.0	22.2	86.7	28.6	54.9	30.0	78.7	33.9	39.1	$\uparrow 3.8$	$\uparrow 4.6$
	Token	601	1158	894	8576	978	8278	888	1568	1049	996	643	1924	$+12.2\%$	$+4.2\%$
Minimality	Acc.	77.1	85.8	30.0	86.7	22.2	83.3	27.6	62.6	26.8	83.0	32.0	39.0	$\uparrow 3.1$	$\uparrow 6.9$
	Token	605	1204	920	9873	1052	9558	849	2528	1158	1275	575	1875	$+14.6\%$	$+21.9\%$
Separability	Acc.	74.7	82.2	25.6	73.3	17.8	65.6	25.6	52.5	30.9	78.7	31.1	31.6	$\uparrow 1.4$	$\downarrow 2.5$
	Token	564	1191	838	7039	923	6940	758	1940	1116	1189	1650	1280	$+29.9\%$	-9.3%
Stability	Acc.	74.0	85.6	27.8	80.0	16.7	80.0	26.8	62.6	31.0	75.0	35.1	34.2	$\uparrow 2.3$	$\uparrow 3.1$
	Token	551	1105	881	7588	976	6497	785	1908	1205	941	781	1215	$+15.0\%$	-10.8%
REST (ours), COMPOSITION OF PROPERTIES															
Best pair	Acc.	77.6	87.0	25.6	83.3	15.6	86.7	29.0	60.1	30.1	85.0	33.9	41.7	$\uparrow 2.4$	$\uparrow 7.5$
	Token	604	1227	916	10281	978	9828	886	2629	1100	1259	613	2134	$+13.2\%$	$+26.7\%$
All properties	Acc.	76.2	86.6	28.9	78.9	21.1	86.7	27.6	63.6	28.7	81.7	33.3	42.1	$\uparrow 3.1$	$\uparrow 6.8$
	Token	608	1253	900	11033	1022	11275	879	2673	1042	1373	569	2151	$+11.5\%$	$+37.9\%$

Table 3 reports the full planner, refiner, and solver system. Every setting improves accuracy over CE-only in the Light system. This trend continues for the Scaled system as well except separability, where it reduces the number of tokens instead. In the Light system, REST with causality or minimality matches or surpasses the strongest frozen agent in Table 6 on all three mathematical benchmarks, while the CE-only system underperforms it on all three. Unlike in the Light single-agent setting, REST increases tokens in the multi-agent system (more in Section 4.4). Beyond the best configuration, the $r = 1$ sweep in Table 11 shows that REST does not rely on careful weight tuning, as both causality and minimality improve the average accuracy at every β .

4.4 Analysis

Scale and Recursion Depth. Figure 4 averages the change in accuracy over CE-only at each recursion round. Additional rounds increase the gain of REST for the Scaled agents and reduce it for the Light agents, in both the single-agent and the multi-agent systems. Larger agents therefore make use of additional rounds of latent exchange, whereas smaller agents obtain their gain from a single round. Appendix D.2 provides the full tables at both rounds.

Auxiliary-Loss Baselines. Table 4 compares REST against CODI and SIM-CoT adapted as loss terms. REST retains an advantage over the two in accuracy on both systems, and in token overhead on the Light system. Both baselines supervise the latent state against a text the producer would have written, whereas REST constrains a property of the thought. CODI requires the consumer’s hidden state to match the state observed under text, while causality requires only that the consumer predict the same continuation. The comparison isolates the effect of an auxiliary term that targets the representation.

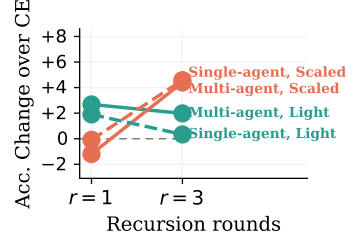


Figure 4. Acc. Change vs r

Table 4. REST against auxiliary-loss baselines, Light vs Scaled.

Method	Metric	Math500		AIME2025		AIME2026		GPQA-D		MedQA		Code Gen.		Avg. Change	
		Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled
CE only	Acc.	71.1	87.7	22.2	76.7	17.8	86.7	24.9	63.1	28.9	80.2	32.5	39.0	Base	Base
	Token	550	1018	849	9009	884	8488	749	2379	868	1137	604	1507	Base	Base
CODI ($\beta = 20$)	Acc.	76.9	85.9	30.0	83.3	22.2	83.3	28.8	61.8	27.3	80.3	32.0	39.5	$\uparrow 3.3$	$\uparrow 0.1$
	Token	598	1035	933	9344	987	9153	861	2293	1101	1097	682	1602	+14.6%	+4.2%
SIM-CoT ($\lambda_{\text{step}} = 0.3$)	Acc.	76.7	82.0	27.8	77.8	21.1	78.9	26.9	54.5	28.9	79.0	34.9	35.0	$\uparrow 3.2$	$\downarrow 4.4$
	Token	598	728	929	9854	1004	8797	914	943	1202	624	549	1276	+15.4%	-5.6%
REST (ours)	Acc.	77.8	88.0	27.8	83.3	22.2	90.0	28.6	67.2	30.0	81.3	33.9	42.3	$\uparrow 3.8$	$\uparrow 3.1$
	Token	601	1044	894	10309	978	9956	888	2287	1049	1120	643	2036	+12.2%	+13.7%

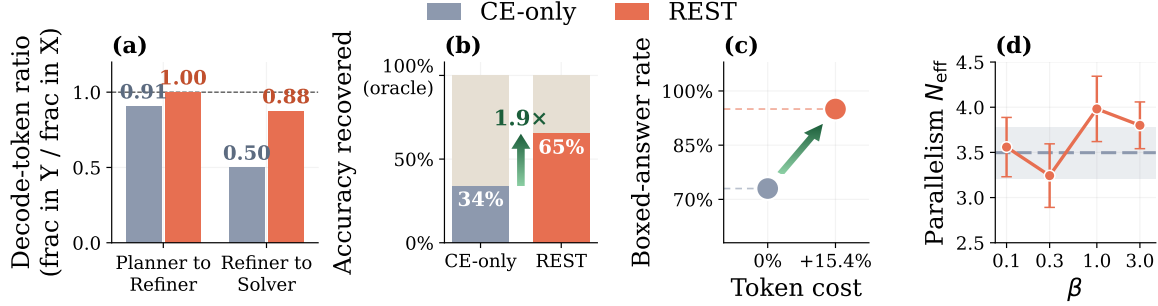


Figure 5. REST against CE-only. (a) Decoded thoughts compared to output / input. (b) Replacing oracle text with thought. (c) Answer rate vs token usage. (d) Effective Superposition.

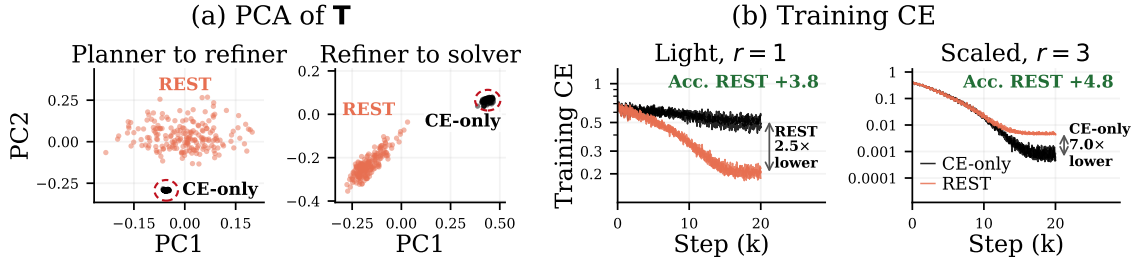


Figure 6. REST against CE-only. (a) PCA for $\mathbf{T}$. CE thoughts collapse into dense clusters, while REST spreads thoughts apart. (b) Training CE under causality. More details in Appendix D.

Encoding Producer Output. Figure 5(a) measures how much of a thought’s decoded content belongs to the producer’s own output, against how much belongs to the prompt it received. Under REST the thought encodes the producer’s plan, or the refined plan. CE has relatively more content from the input encoded, which might be irrelevant to the next agent. This is exactly the behavior targeted by eq. (3). The effect grows as the system moves further along the chain of latent thoughts.

Preserved Information. Figure 5(b) compares the performance of the solver when given the oracle refined plan compared to the thought representation. REST reaches more of this accuracy than CE. Therefore, REST contains relatively more useful information to the solver than CE.

Token Cost. This information comes at a token cost. Although we observe an increase in tokens when using REST (+15.4% on average), this leads to the model being more likely to generate a final answer. We analyze the boxed-answer rate which is the percentage of examples the model converges on an answer. As illustrated in Figure 5(c), REST reaches a 95% boxed-answer rate compared to 73% for CE. This indicates that the additional tokens are spent towards reaching a final answer.

Superposition. Figure 5(d) measures how many candidate reasoning paths a thought supports at once, a metric we adapt from Deng et al. [13] (eq. (78)). It decodes $\mathbf{T}$ through the consumer’s vocabulary, scores each candidate refined plan by the decoded tokens it receives, and reports the exponentiated entropy of the resulting posterior over candidates. Higher N_{eff} indicates higher effective superposition. REST maintains superposition and slightly increases with increasing the weight β .

CE Thoughts. Figure 6(a) projects $\mathbf{T}$ for 200 random samples. Thoughts from CE are gathered into dense clusters, hence semantically distinct questions would have close representations that would confuse a consumer agent. After inspection of the CE-only clusters, the texts are not related to each other (Appendix D). Therefore, the clustering represents collapse with no shared content. REST penalizes this behavior through separability across a continuous region. This is the failure in Section 2.1, where difficult examples would be indistinguishable from a collapse. Lower training CE does not achieve better accuracy (Figure 6(b)). CE-only plateaus above causality for Light, where it does not memorize, while for Scaled it memorizes the training set and still underperforms REST in accuracy. Therefore, only minimizing CE would make the thought representation underconstrained.

5 Related Work

Reasoning in Latent Space. Recent work treats the continuous latent space of an LLM as the medium of reasoning rather than the decoded text [6, 56]. Within a single model, hidden states re-enter the input stream as the next reasoning step [2, 27, 41, 43, 53]. Across models, hidden representations and KV caches carry information between agents in place of text [15, 31, 47, 50, 54, 59]. A further line reuses the same computation over several rounds to deepen latent reasoning [1, 16, 24, 57], which Zou et al. [58] raise to the system level by looping heterogeneous agents through a learned link. Most of these systems rely on the cross-entropy of the final decoded answer for training, and typically leave the latent thoughts themselves unconstrained.

Supervising Latent Thoughts. The signal that trains a latent state is the answer following the latent steps [19], a self-distilled explicit chain of thought [40], or an auxiliary decoder aligning every latent step with its explicit counterpart [46]. Each of these signals is defined against a reference the latent state must reproduce, either a text or another model’s own cache [4]. Audits of the resulting representations report high accuracy without latent reasoning, collapsed candidate solutions, and unstable trajectories [11, 37, 38]. A recent framework states four properties a valid thought representation should satisfy, and audits existing methods against them [39]. REST is the first to turn the theoretically motivated properties of a valid thought representation into loss terms for a latent recursive LLM system (Figure 7). Additional related work in Appendix C.

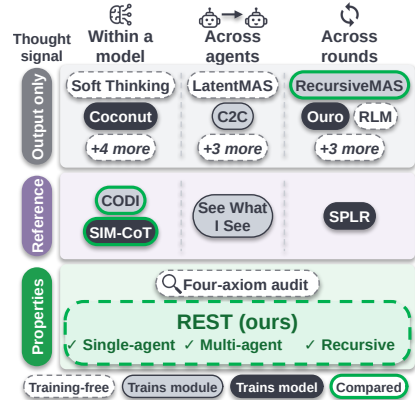


Figure 7. Positioning of REST.

6 Conclusion

We introduce REST, a training objective that supervises the thoughts of latent recursive LLM systems through four properties of a valid thought representation. Our objective translates causality, minimality, separability, and stability into differentiable loss terms added to CE. These terms require no architectural changes or added parameters at inference. Theoretical analyses show that the failures these properties prevent lead to a lower probability of the correct answer. Empirically, thoughts trained only with CE collapse across distinct questions and retain irrelevant information. Evaluations across mathematical, scientific, medical, and code-generation benchmarks demonstrate that REST improves accuracy over CE in both single- and multi-agent settings. Overall, REST establishes a principled way to supervise latent thoughts beyond the final decoded answer.

Implications. The improvements from an auxiliary term in both settings indicate that CE-only insufficiently constrains latent thoughts, and that latent recursive systems benefit from supervising the thought representation. The property that yields the largest gain further depends on the consumer, as causality leads when the thought passes to a different agent and minimality leads when a model recurs on its own thought. Since the collapse and input content observed under CE are not reflected in the CE objective, final-answer accuracy provides an incomplete view of latent systems, and inspection of the thought representation should accompany it. Finally, decoded REST thoughts contain relatively more of the producer’s output than of its prompt, suggesting that property supervision can make latent communication easier to audit as agents increasingly communicate outside of text.

Limitations. Due to the substantial computational cost, we do not sweep all combinations of properties and hyperparameters, as the number of required runs grows multiplicatively with each addition. Nevertheless, causality and minimality improve over CE at every weight we evaluate (Table 11). A cheap approximation to Stability is opted for due to compute limitations but still leads to performance gain. Another limitation is that REST trains only the outer link while the base LLMs and the inner link remain frozen, a design choice to isolate the contribution of each property term. Extending REST to the training of the inner link or the agent is left to future work.

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A Theoretical Analysis

A.1 From Axioms to Differentiable Surrogates

This subsection derives each differentiable loss term used in our work from the functional properties of [39]. Starting from the definition of each property, we arrive at the loss term used in Section 3.

Causality

Chaining the per-position distributions of Section 3.1 yields the distribution that the consumer assigns to a complete target,

$$P(v | Z) = \prod_{t=1}^n p(v_t | v_{<t}, Z), \quad (8)$$

where truncating the product at $t-1$ gives the distribution $P(v_{<t} | Z)$ it assigns to a prefix. Only $p(\cdot | v_{<t}, \mathbf{T})$ and $P(\cdot | \mathbf{T})$ depend on ψ , since a textual transfer consists of embeddings independent of ψ and every θ_i and $\mathcal{R}_{\text{in}}$ are frozen.

In the notation of eq. (8), the Causality axiom requires that both transfers induce the same distribution over complete targets, and the divergence

$$D_{\text{KL}}(P(\cdot | E(u)) \| P(\cdot | \mathbf{T})) \quad (9)$$

quantifies the extent to which they do not. Evaluating eq. (9) ranges over every continuation in $\mathcal{V}^n$ and therefore requires autoregressive sampling, whereas one teacher-forced pass yields a single continuation at all n positions simultaneously. Equation (2) is eq. (9) in differentiable form, and two differences separate them. The surrogate averages a per-position divergence, and it takes its prefixes from the dataset. This is contrary to a divergence over sequences and to the continuations a textual transfer would produce. Since we can factorize the divergence, the per-position average does not introduce error.

Proposition A.1 (Exact factorization). *For every transfer,*

$$D_{\text{KL}}(P(\cdot | E(u)) \| P(\cdot | \mathbf{T})) = \sum_{t=1}^n \mathbb{E}_{v_{<t} \sim P(\cdot | E(u))} \left[D_{\text{KL}}(p(\cdot | v_{<t}, E(u)) \| p(\cdot | v_{<t}, \mathbf{T})) \right]. \quad (10)$$

Proof of Proposition A.1. Expanding the divergence over complete targets,

$$\begin{aligned} D_{\text{KL}}(P(\cdot | E(u)) \| P(\cdot | \mathbf{T})) &= \sum_{v \in \mathcal{V}^n} P(v | E(u)) \log \frac{P(v | E(u))}{P(v | \mathbf{T})} \\ &\stackrel{\text{equation 8}}{=} \sum_{v \in \mathcal{V}^n} P(v | E(u)) \sum_{t=1}^n \log \frac{p(v_t | v_{<t}, E(u))}{p(v_t | v_{<t}, \mathbf{T})}, \end{aligned}$$

where the second line writes the logarithm of a product as a sum of logarithms. Both sums are finite, hence exchanging them gives

$$D_{\text{KL}}(P(\cdot | E(u)) \| P(\cdot | \mathbf{T})) = \sum_{t=1}^n \sum_{v \in \mathcal{V}^n} P(v | E(u)) \log \frac{p(v_t | v_{<t}, E(u))}{p(v_t | v_{<t}, \mathbf{T})}. \quad (11)$$

Fix a position t and evaluate its inner sum. Its summand depends on v only through $v_{\leq t}$. Therefore, summing over every continuation past t reduces it to a sum over $\mathcal{V}^t$. The prefix factorization

$$P(v_{\leq t} | E(u)) = P(v_{<t} | E(u)) p(v_t | v_{<t}, E(u)) \quad (12)$$

then splits that sum over the prefix $v_{<t}$ and the final token $v_t = w$,

$$\begin{aligned}
 & \sum_{v \in \mathcal{V}^n} P(v \mid E(u)) \log \frac{p(v_t \mid v_{<t}, E(u))}{p(v_t \mid v_{<t}, \mathbf{T})} \\
 &= \sum_{v_{\leq t} \in \mathcal{V}^t} P(v_{\leq t} \mid E(u)) \log \frac{p(v_t \mid v_{<t}, E(u))}{p(v_t \mid v_{<t}, \mathbf{T})} \\
 &= \sum_{v_{<t} \in \mathcal{V}^{t-1}} P(v_{<t} \mid E(u)) \sum_{w \in \mathcal{V}} p(w \mid v_{<t}, E(u)) \log \frac{p(w \mid v_{<t}, E(u))}{p(w \mid v_{<t}, \mathbf{T})} \\
 &= \sum_{v_{<t} \in \mathcal{V}^{t-1}} P(v_{<t} \mid E(u)) D_{\text{KL}}(p(\cdot \mid v_{<t}, E(u)) \parallel p(\cdot \mid v_{<t}, \mathbf{T})) \\
 &= \mathbb{E}_{v_{<t} \sim P(\cdot \mid E(u))} \left[D_{\text{KL}}(p(\cdot \mid v_{<t}, E(u)) \parallel p(\cdot \mid v_{<t}, \mathbf{T})) \right],
 \end{aligned}$$

where the third equality recognizes the inner sum over w as a divergence. The fourth recognizes the outer sum as an expectation over the prefixes a textual transfer produces. Substituting this equality into eq. (11) at every position yields eq. (10).

The second difference therefore remains, which is the distribution the prefixes $v_{<t}$ of eq. (10) are sampled from. This is the only step at which the surrogate is not the exact property, since training samples those prefixes from the dataset rather than from the consumer’s own generations. Two conditions bound its effect.

Assumption A.1 (Bounded disagreement). There exists a finite B with

$$\left| \log p(w \mid v_{<t}, E(u)) - \log p(w \mid v_{<t}, \mathbf{T}) \right| \leq B \quad (13)$$

at every position t , every prefix $v_{<t}$, every token $w \in \mathcal{V}$, and every ψ in the optimization domain.

Bounding the consumer’s logits z in absolute value by M lower-bounds every next-token probability,

$$p(w \mid v_{<t}, Z) = \frac{\exp(z_w)}{\sum_{w' \in \mathcal{V}} \exp(z_{w'})} \geq \frac{e^{-M}}{|\mathcal{V}| e^M} = \frac{e^{-2M}}{|\mathcal{V}|},$$

hence every log-probability lies in an interval of length

$$\log 1 - \log \frac{e^{-2M}}{|\mathcal{V}|} = 2M + \log |\mathcal{V}|.$$

Every difference of two of them therefore satisfies

$$\left| \log p(w \mid v_{<t}, E(u)) - \log p(w \mid v_{<t}, \mathbf{T}) \right| \leq 2M + \log |\mathcal{V}|,$$

at which Assumption A.1 holds with $B = 2M + \log |\mathcal{V}|$.¹ Equation (13) bounds the corresponding divergence by the same constant,

$$D_{\text{KL}}(p(\cdot \mid v_{<t}, E(u)) \parallel p(\cdot \mid v_{<t}, \mathbf{T})) = \sum_w p(w \mid v_{<t}, E(u)) \log \frac{p(w \mid v_{<t}, E(u))}{p(w \mid v_{<t}, \mathbf{T})} \leq B, \quad (14)$$

where the inequality averages eq. (13) under $p(\cdot \mid v_{<t}, E(u))$.

Assumption A.2 (Reference faithfulness). Let D denote the distribution from which the training data draws the target v , conditioned on the same E_{pre} and E_{post} that the consumer receives in Definition 2.1. There exists $\varepsilon \in [0, 1]$ with $d_{\text{TV}}(D, P(\cdot \mid E(u))) \leq \varepsilon$.

¹This constant is the worst case over the whole vocabulary.

Assumption A.2 requires that the consumer, once given the producer’s text explicitly, reproduce what the training data records. The training trajectory in general comes from a different model than the frozen consumer, and ε therefore quantifies a distillation gap between the two.

We here state a standard result on averaging a bounded quantity under two distributions, for completeness. It will later let us replace the distribution that weights each position.

Lemma A.1 (Total variation controls a bounded average). *Averaging a quantity whose range fits in an interval of length B under two distributions π and π' on a common finite space alters the value by at most B times their total variation distance,*

$$|\mathbb{E}_\pi[f] - \mathbb{E}_{\pi'}[f]| \leq B \cdot d_{\text{TV}}(\pi, \pi'). \quad (15)$$

Proof of Lemma A.1. Let c denote the midpoint of the interval containing f ’s range, at which

$$|f - c| \leq \frac{B}{2}.$$

Then

$$|\mathbb{E}_\pi[f] - \mathbb{E}_{\pi'}[f]| = \left| \sum (\pi - \pi')(f - c) \right| \leq \frac{B}{2} \sum |\pi - \pi'| = B \cdot d_{\text{TV}}(\pi, \pi'),$$

where subtracting c leaves the difference unchanged because $\sum (\pi - \pi') = 0$.

Proposition A.2 (The surrogate tracks the axiom). *Under Assumptions A.1 and A.2, for every ψ in the optimization domain, the surrogate averaged over the training target,*

$$\mathbb{E}_{v \sim D} [\mathcal{L}_{\text{caus}}(\psi)] = \mathbb{E}_{v \sim D} \left[\frac{1}{n} \sum_{t=1}^n D_{\text{KL}}(p(\cdot | v_{<t}, E(u)) \| p(\cdot | v_{<t}, \mathbf{T})) \right],$$

satisfies

$$\left| \mathbb{E}_{v \sim D} [\mathcal{L}_{\text{caus}}(\psi)] - \frac{1}{n} D_{\text{KL}}(P(\cdot | E(u)) \| P(\cdot | \mathbf{T})) \right| \leq B\varepsilon. \quad (16)$$

Proof of Proposition A.2. Denote the divergence at a single position by

$$\Delta_t(v_{<t}) = D_{\text{KL}}(p(\cdot | v_{<t}, E(u)) \| p(\cdot | v_{<t}, \mathbf{T})) \in [0, B],$$

where the lower end holds of every divergence and the upper end follows from eq. (14). In this notation eq. (2) expands to

$$\mathcal{L}_{\text{caus}}(\psi) = \frac{1}{n} \sum_{t=1}^n D_{\text{KL}}(p(\cdot | v_{<t}, E(u)) \| p(\cdot | v_{<t}, \mathbf{T})) = \frac{1}{n} \sum_{t=1}^n \Delta_t(v_{<t}),$$

and Proposition A.1 becomes

$$\frac{1}{n} D_{\text{KL}}(P(\cdot | E(u)) \| P(\cdot | \mathbf{T})) = \frac{1}{n} \sum_{t=1}^n \mathbb{E}_{P(\cdot | E(u))} [\Delta_t],$$

where both are expressed through the same Δ_t and differ only in how each position is weighted. The two weightings are the distributions of $v_{<t}$ under D and under $P(\cdot | E(u))$, each obtained through the projection

$$\pi_t : \mathcal{V}^n \rightarrow \mathcal{V}^{t-1}, \quad \pi_t(v) = v_{<t}.$$

Projecting cannot increase total variation distance,

$$\begin{aligned} d_{\text{TV}}(D \circ \pi_t^{-1}, P(\cdot | E(u)) \circ \pi_t^{-1}) \\ &\leq d_{\text{TV}}(D, P(\cdot | E(u))) && \text{data processing} \\ &\leq \varepsilon && \text{Assumption A.2,} \end{aligned}$$

and Lemma A.1 with $f = \Delta_t$ yields

$$\left| \mathbb{E}_D[\Delta_t] - \mathbb{E}_{P(\cdot|E(u))}[\Delta_t] \right| \leq B\varepsilon \quad \text{at every position } t.$$

Summing the n positions and applying the triangle inequality,

$$\begin{aligned} & \left| n \mathbb{E}_{v \sim D}[\mathcal{L}_{\text{caus}}(\psi)] - D_{\text{KL}}(P(\cdot | E(u)) \| P(\cdot | \mathbf{T})) \right| \\ &= \left| \sum_{t=1}^n \left(\mathbb{E}_D[\Delta_t] - \mathbb{E}_{P(\cdot|E(u))}[\Delta_t] \right) \right| \\ &\leq \sum_{t=1}^n \left| \mathbb{E}_D[\Delta_t] - \mathbb{E}_{P(\cdot|E(u))}[\Delta_t] \right| \leq nB\varepsilon, \end{aligned}$$

and dividing by n yields eq. (16).

The division by n in eq. (2) renders eq. (16) independent of the target length. The n per-position bounds accumulate to $nB\varepsilon$, and the normalization then removes the factor. Within a transfer that division is a constant free of ψ , hence the minimizer is the same either way. Proposition A.2 compares the two quantities, and the following corollary establishes that minimizing one minimizes the other up to a bounded excess.

Corollary A.1 (Surrogate minimizers are near-optimal for the axiom). *Under Assumptions A.1 and A.2, let $\hat{\psi}$ minimize $\mathbb{E}_{v \sim D}[\mathcal{L}_{\text{caus}}(\psi)]$ over the optimization domain and let ψ^* minimize eq. (9) over the same domain. Then eq. (9) at $\hat{\psi}$ exceeds its minimum by at most $2nB\varepsilon$.*

Proof of Corollary A.1. Denote the two objectives by

$$F(\psi) = \frac{1}{n} D_{\text{KL}}(P(\cdot | E(u)) \| P(\cdot | \mathbf{T})), \quad G(\psi) = \mathbb{E}_{v \sim D}[\mathcal{L}_{\text{caus}}(\psi)],$$

where $|F(\psi) - G(\psi)| \leq B\varepsilon$ at every ψ by Proposition A.2. Chaining the two objectives at $\hat{\psi}$ and at ψ^* ,

$$\begin{aligned} F(\hat{\psi}) &\leq G(\hat{\psi}) + B\varepsilon && \text{Proposition A.2} \\ &\leq G(\psi^*) + B\varepsilon && \hat{\psi} \text{ minimizes } G \\ &\leq F(\psi^*) + 2B\varepsilon && \text{Proposition A.2.} \end{aligned}$$

Multiplying by n ,

$$nF(\hat{\psi}) \leq nF(\psi^*) + 2nB\varepsilon,$$

where $nF(\psi)$ is eq. (9).

Minimality

The Minimality axiom requires that the thought minimize the Information Bottleneck Lagrangian at a trade-off weight β_{IB} [39], formed from the mutual information $I(\cdot; \cdot)$. This weight is internal to the Minimality derivation and is distinct from the β of eq. (7), which weights a property term against the CE objective.

$$J(\beta_{\text{IB}}) = I(X; \mathbf{T}) - \beta_{\text{IB}} I(\mathbf{T}; Y), \quad (17)$$

retaining little of the input and much of the output. Evaluating eq. (17) requires distributions that training would have to estimate, whereas each cross-entropy of eq. (3) follows from one forward pass of the consumer. The two weights of eq. (3) enter the results below only through

$$\beta_{\text{IB}} = 1 + \frac{\lambda_1}{\lambda_2}, \quad \text{equivalently} \quad \lambda_2(\beta_{\text{IB}} - 1) = \lambda_1. \quad (18)$$

The ratio λ_1/λ_2 covers $(0, \infty)$, hence eq. (18) covers every $\beta_{\text{IB}} > 1$. Therefore, eq. (3) represents eq. (17) at every such β_{IB} , with that ratio setting the trade-off weight. The symmetric weight $\beta_{\text{IB}} = 2$ corresponds to $\lambda_1 = \lambda_2$.

Equation (3) is eq. (17) in differentiable form. Proposition A.3 rewrites the axiom in entropies, Proposition A.4 replaces those entropies by the cross-entropies the consumer incurs, and the two conditions that follow bound the residual terms of that replacement.

Proposition A.3 (Decomposition of the axiom). *Under eq. (18),*

$$\lambda_2 J(\beta_{\text{IB}}) = \lambda_1 \mathbb{H}(Y | \mathbf{T}) - \lambda_2 \mathbb{H}(X | Y, \mathbf{T}) - \lambda_2 I(Y; \mathbf{T} | X) + C, \quad (19)$$

where $C = \lambda_2 \mathbb{H}(X | Y) - \lambda_1 \mathbb{H}(Y)$ is independent of $\mathbf{T}$.

Proof of Proposition A.3. The chain rule for mutual information accumulates the information $\mathbf{T}$ contains about the pair (X, Y) in either order,

$$I(X; \mathbf{T}) + I(Y; \mathbf{T} | X) = I(\mathbf{T}; Y) + I(X; \mathbf{T} | Y),$$

and solving for $I(X; \mathbf{T})$ and substituting into eq. (17) gives

$$J(\beta_{\text{IB}}) = (1 - \beta_{\text{IB}}) I(\mathbf{T}; Y) + I(X; \mathbf{T} | Y) - I(Y; \mathbf{T} | X).$$

Multiplying by $\lambda_2 > 0$ and applying $\lambda_2(\beta_{\text{IB}} - 1) = \lambda_1$ to the first term,

$$\lambda_2 J(\beta_{\text{IB}}) = -\lambda_1 I(\mathbf{T}; Y) + \lambda_2 I(X; \mathbf{T} | Y) - \lambda_2 I(Y; \mathbf{T} | X).$$

The first two expand into entropies,

$$I(\mathbf{T}; Y) = \mathbb{H}(Y) - \mathbb{H}(Y | \mathbf{T}), \quad I(X; \mathbf{T} | Y) = \mathbb{H}(X | Y) - \mathbb{H}(X | Y, \mathbf{T}),$$

and substituting them gives

$$\lambda_2 J(\beta_{\text{IB}}) = \lambda_1 \mathbb{H}(Y | \mathbf{T}) - \lambda_2 \mathbb{H}(X | Y, \mathbf{T}) - \lambda_2 I(Y; \mathbf{T} | X) + \underbrace{\lambda_2 \mathbb{H}(X | Y) - \lambda_1 \mathbb{H}(Y)}_C.$$

The axiom's original decomposition discards $I(Y; \mathbf{T} | X)$, which equals zero when $\mathbf{T}$ is only a function of X . Proposition A.3 retains it, since $\mathbf{T} = \mathcal{R}_\psi(\mathcal{R}_{\text{in}}(H_u))$ takes the hidden states of a pass over the producer's output as its input and therefore depends on Y .

Proposition A.4 (The axiom in terms of the loss). *Let $\gamma_1 = \text{CE}(Y | \mathbf{T}) - \mathbb{H}(Y | \mathbf{T})$ and $\gamma_2 = \text{CE}(X | Y, \mathbf{T}) - \mathbb{H}(X | Y, \mathbf{T})$ denote the two decoder gaps. Then under eq. (18),*

$$\begin{aligned} \lambda_2 J(\beta_{\text{IB}}) = & \underbrace{\lambda_1 \text{CE}(Y | \mathbf{T}) - \lambda_2 \text{CE}(X | Y, \mathbf{T})}_{\mathcal{L}_{\min}(\psi) \text{ of eq. (3)}} + C \\ & - \lambda_2 I(Y; \mathbf{T} | X) - (\lambda_1 \gamma_1 - \lambda_2 \gamma_2). \end{aligned} \quad (20)$$

Proof of Proposition A.4. Gibbs' inequality lower-bounds a cross-entropy by the entropy it approximates,

$$\text{CE}(Y | \mathbf{T}) \geq \mathbb{H}(Y | \mathbf{T}), \quad \text{CE}(X | Y, \mathbf{T}) \geq \mathbb{H}(X | Y, \mathbf{T}),$$

hence $\gamma_1 \geq 0$ and $\gamma_2 \geq 0$, and their definitions rearrange to

$$\mathbb{H}(Y | \mathbf{T}) = \text{CE}(Y | \mathbf{T}) - \gamma_1, \quad \mathbb{H}(X | Y, \mathbf{T}) = \text{CE}(X | Y, \mathbf{T}) - \gamma_2.$$

Substituting both into the entropy pair of eq. (19),

$$\begin{aligned} \lambda_1 \mathbb{H}(Y | \mathbf{T}) - \lambda_2 \mathbb{H}(X | Y, \mathbf{T}) &= \lambda_1 (\text{CE}(Y | \mathbf{T}) - \gamma_1) - \lambda_2 (\text{CE}(X | Y, \mathbf{T}) - \gamma_2) \\ &= \lambda_1 \text{CE}(Y | \mathbf{T}) - \lambda_2 \text{CE}(X | Y, \mathbf{T}) - (\lambda_1 \gamma_1 - \lambda_2 \gamma_2). \end{aligned}$$

Equation (19) gives the same entropy pair as

$$\lambda_1 \mathbb{H}(Y | \mathbf{T}) - \lambda_2 \mathbb{H}(X | Y, \mathbf{T}) = \lambda_2 J(\beta_{\text{IB}}) + \lambda_2 I(Y; \mathbf{T} | X) - C,$$

and equating the two right sides and solving for $\lambda_2 J(\beta_{\text{IB}})$ gives eq. (20).

Equation (20) is exact, and two conditions bound the terms by which eq. (3) exceeds the axiom's objective.

Assumption A.3 (Bounded decoder gap). There exists a finite Γ with $|\lambda_1 \gamma_1 - \lambda_2 \gamma_2| \leq \Gamma$ at every ψ in the optimization domain.

Gibbs' inequality constrains the first gap,

$$\gamma_1 = \text{CE}(Y | \mathbf{T}) - \mathbb{H}(Y | \mathbf{T}) \geq 0,$$

and minimizing eq. (3) decreases γ_1 toward zero once the decoder approaches the true conditional distribution of Y given $\mathbf{T}$. The second gap admits no such bound. The coefficient of $\text{CE}(X | Y, \mathbf{T})$ in eq. (3) is $-\lambda_2$, hence minimizing the surrogate increases that cross-entropy. Any upper bound on γ_2 derived from that cross-entropy increases as well, while $\mathbb{H}(X | Y, \mathbf{T})$ remains unconstrained. Assumption A.3 therefore requires the consumer to decode with comparable quality under both prompts of eq. (3). Rearranging it,

$$\gamma_1 \geq \frac{\lambda_2 \gamma_2 - \Gamma}{\lambda_1},$$

constrains γ_2 by γ_1 at every ψ . Equation (18) turns the ordering of the two weights into an ordering of β_{IB} ,

$$\lambda_2 < \lambda_1 \iff \frac{\lambda_1}{\lambda_2} > 1 \iff \beta_{\text{IB}} > 2,$$

at which the constrained direction has the larger weight.

Assumption A.4 (Bounded realization leakage). There exists a finite κ with $I(Y; \mathbf{T} | X) \leq \kappa$ at every ψ in the optimization domain.

Assumption A.4 bounds what $\mathbf{T}$ retains of the particular output beyond what the input explains, and it holds with $\kappa = 0$ when $\mathbf{T}$ is only a function of X . Solving eq. (20) for the surrogate,

$$\mathcal{L}_{\min}(\psi) = \lambda_2 J(\beta_{\text{IB}}) - C + \lambda_2 I(Y; \mathbf{T} | X) + (\lambda_1 \gamma_1 - \lambda_2 \gamma_2), \quad (21)$$

gives the term a positive coefficient, hence minimizing eq. (3) reduces it alongside the axiom's objective rather than opposed to it.

Proposition A.5 (The surrogate tracks the axiom). Under Assumptions A.3 and A.4, at every ψ in the optimization domain,

$$\left| \lambda_2 J(\beta_{\text{IB}}) - (\mathcal{L}_{\min}(\psi) + C) \right| \leq \Gamma + \lambda_2 \kappa. \quad (22)$$

Proof of Proposition A.5. Rearranging eq. (20),

$$\lambda_2 J(\beta_{\text{IB}}) - (\mathcal{L}_{\min}(\psi) + C) = -\lambda_2 I(Y; \mathbf{T} | X) - (\lambda_1 \gamma_1 - \lambda_2 \gamma_2),$$

and bounding the two terms in turn,

$$\begin{aligned} & \left| \lambda_2 J(\beta_{\text{IB}}) - (\mathcal{L}_{\min}(\psi) + C) \right| \\ & \leq \lambda_2 \left| I(Y; \mathbf{T} | X) \right| + \left| \lambda_1 \gamma_1 - \lambda_2 \gamma_2 \right| && \text{triangle inequality} \\ & \leq \lambda_2 \kappa + \Gamma && \text{Assumptions A.3 and A.4.} \end{aligned}$$

Proposition A.5 compares the two quantities, and the following corollary establishes that minimizing one minimizes the other up to a bounded excess.

Corollary A.2 (Surrogate minimizers are near-optimal for the axiom). *Under Assumptions A.3 and A.4, let $\hat{\psi}$ minimize eq. (3) over the optimization domain and let ψ^* minimize eq. (17) over the same domain. Then*

$$J(\beta_{\text{IB}})|_{\hat{\psi}} \leq J(\beta_{\text{IB}})|_{\psi^*} + \frac{2(\Gamma + \lambda_2 \kappa)}{\lambda_2}. \quad (23)$$

Proof of Corollary A.2. Denote the two objectives by

$$F(\psi) = \lambda_2 J(\beta_{\text{IB}}), \quad G(\psi) = \lambda_1 \text{CE}(Y | \mathbf{T}) - \lambda_2 \text{CE}(X | Y, \mathbf{T}) + C,$$

where $|F(\psi) - G(\psi)| \leq \Gamma + \lambda_2 \kappa$ at every ψ by Proposition A.5. Since C and λ_2 are constant in ψ , $\hat{\psi}$ minimizes G and ψ^* minimizes F . Similar to how we showed Corollary A.1, we chain the two objectives at $\hat{\psi}$ and at ψ^* to obtain

$$F(\hat{\psi}) \leq F(\psi^*) + 2(\Gamma + \lambda_2 \kappa).$$

Substituting $F(\psi) = \lambda_2 J(\beta_{\text{IB}})$ and dividing by $\lambda_2 > 0$,

$$J(\beta_{\text{IB}})|_{\hat{\psi}} \leq J(\beta_{\text{IB}})|_{\psi^*} + \frac{2(\Gamma + \lambda_2 \kappa)}{\lambda_2}.$$

Separability

The Separability axiom requires a projection of bounded capacity under which the thoughts of two semantically disjoint inputs stay apart,

$$d(\varphi(\mathbf{T}), \varphi(\mathbf{T}')) > \delta \quad \text{for some } \varphi \in \mathcal{H}, \quad (24)$$

where $\mathcal{H}$ is the class of admissible projections and d is a metric on the semantic space. Normalizing the pooled thought of eq. (4) to unit length,

$$\varphi_\omega(\mathbf{T}) = \mathbf{t} / \|\mathbf{t}\|, \quad (25)$$

gives a member of such a class. Expanding the squared distance between two of its values,

$$\begin{aligned} \|\varphi_\omega(\mathbf{T}) - \varphi_\omega(\mathbf{T}')\|^2 &= \|\varphi_\omega(\mathbf{T})\|^2 + \|\varphi_\omega(\mathbf{T}')\|^2 - 2\langle \varphi_\omega(\mathbf{T}), \varphi_\omega(\mathbf{T}') \rangle \\ &= 2 - 2 \frac{\langle \mathbf{t}, \mathbf{t}' \rangle}{\|\mathbf{t}\| \|\mathbf{t}'\|} \\ &= 2 - 2\text{sim}(\mathbf{t}, \mathbf{t}'), \end{aligned} \quad (26)$$

where the second equality applies the unit norm that eq. (25) imposes on both values. We adopt the distance of eq. (26) as the metric d of eq. (24), following the realization of the semantic metric by cosine similarity in Seddik and Fard [39]. Equation (4) is eq. (24) in differentiable form. The axiom asserts an inequality, hence the two are linked by a threshold, and the axiom is valid whenever the surrogate falls below it.

Proposition A.6 (Lower bound on the margin). *For every ψ , every ω , and every preceding thought $\mathbf{T}_k$ with pooled vector $\mathbf{t}_k$, $k \in \{1, \dots, K\}$, as in Section 3.1,*

$$\|\varphi_\omega(\mathbf{T}) - \varphi_\omega(\mathbf{T}_k)\|^2 \geq 2 - 2\tau \mathcal{L}_{\text{sep}}(\psi, \omega) \quad \text{at every } k \in \{1, \dots, K\}, \quad K \geq 1. \quad (27)$$

Proof of Proposition A.6. Every term of the sum in eq. (4) is positive, hence each single term is at most the whole sum,

$$\exp(\text{sim}(\mathbf{t}, \mathbf{t}_k)/\tau) \leq \sum_{j=1}^K \exp(\text{sim}(\mathbf{t}, \mathbf{t}_j)/\tau) \quad \text{at every } k.$$

Taking logarithms,

$$\frac{1}{\tau} \text{sim}(\mathbf{t}, \mathbf{t}_k) \leq \log \sum_{j=1}^K \exp(\text{sim}(\mathbf{t}, \mathbf{t}_j)/\tau) = \mathcal{L}_{\text{sep}}(\psi, \omega),$$

and multiplying by $\tau > 0$,

$$\text{sim}(\mathbf{t}, \mathbf{t}_k) \leq \tau \mathcal{L}_{\text{sep}}(\psi, \omega) \quad \text{at every } k. \quad (28)$$

Substituting eq. (28) into eq. (26),

$$\|\varphi_\omega(\mathbf{T}) - \varphi_\omega(\mathbf{T}_k)\|^2 = 2 - 2\text{sim}(\mathbf{t}, \mathbf{t}_k) \geq 2 - 2\tau \mathcal{L}_{\text{sep}}(\psi, \omega),$$

which is eq. (27).

Equation (28) holds at all K indices at once, hence the surrogate controls the largest similarity rather than their average. The right side of eq. (27) is positive if and only if

$$\tau \mathcal{L}_{\text{sep}}(\psi, \omega) < 1.$$

Equation (24) requires that margin only of inputs whose semantic supports are disjoint, and one condition supplies that precondition at the pairs the surrogate acts on.

Assumption A.5 (Distinct inputs are semantically disjoint). The K preceding thoughts originate from training examples distinct from the current one, and two distinct examples induce disjoint semantic supports.

The semantic support of an example covers its whole reasoning trajectory together with its conclusion, hence two distinct inputs violate Assumption A.5 only when both their trajectories and their conclusions agree. Assumption A.5 makes every pair the surrogate acts on semantically disjoint, hence each pair falls under the clause proved here rather than under the axiom's converse clause, which applies to semantically convergent inputs.

Corollary A.3 (Sufficient condition for the axiom's margin). Under Assumption A.5, at every $\delta \in (0, 2]$,

$$\underbrace{\log \sum_{k=1}^K \exp(\text{sim}(\mathbf{t}, \mathbf{t}_k)/\tau)}_{\mathcal{L}_{\text{sep}}(\psi, \omega) \text{ of eq. (4)}} \leq \frac{2 - \delta^2}{2\tau} \implies \|\varphi_\omega(\mathbf{T}) - \varphi_\omega(\mathbf{T}_k)\| \geq \delta \quad \text{at every } k, \quad (29)$$

and eq. (24) therefore holds at every $\delta' \in (0, \delta)$ with $\varphi = \varphi_\omega$.

Proof of Corollary A.3. Multiplying the hypothesis of eq. (29) by 2τ and rearranging gives

$$2 - 2\tau \mathcal{L}_{\text{sep}}(\psi, \omega) \geq \delta^2,$$

and chaining this with Proposition A.6 lower-bounds every squared distance by δ^2 ,

$$\|\varphi_\omega(\mathbf{T}) - \varphi_\omega(\mathbf{T}_k)\|^2 \geq 2 - 2\tau \mathcal{L}_{\text{sep}}(\psi, \omega) \geq \delta^2 \quad \text{at every } k.$$

Taking square roots yields the conclusion of eq. (29). Assumption A.5 supplies the precondition of eq. (24) at each pair, and φ_ω belongs to $\mathcal{H}$ by eq. (25).

Corollary A.3 states the axiom's own requirement as a threshold on the surrogate, hence minimizing eq. (4) maximizes the margin obtainable from $\mathcal{H}$.

Stability

Chaining the per-position distributions of eq. (5) yields the distribution the producer assigns to a complete output,

$$Q(u) = \prod_{s=1}^m q(u_s | u_{<s}), \quad (30)$$

and Q is the distribution of the producer's output Y . The axiom comprises a lexical-invariance clause and a mode-collapse clause, and eq. (6) targets the latter, which requires that $\mathbf{T}$ encode the entropy

$$\mathbb{H}(Y) = - \sum_{u \in \mathcal{V}^m} Q(u) \log Q(u) \quad (31)$$

rather than a single realization drawn from Q . Evaluating eq. (31) ranges over every output in $\mathcal{V}^m$ and therefore requires autoregressive sampling, whereas one teacher-forced pass yields all m next-token distributions along a single output. Equation (6) is eq. (31) in differentiable form, and two differences separate its target eq. (5) from the axiom. The estimate averages per-position entropies along one output, and it evaluates those entropies along the output in the dataset. This is contrary to an entropy over sequences and to outputs from the producer itself. Since we can factorize the entropy, the per-position average does not introduce error.

Proposition A.7 (Exact factorization). *For every producer,*

$$\mathbb{H}(Y) = \sum_{s=1}^m \mathbb{E}_{u_{<s} \sim Q} \left[\mathbb{H}(q(\cdot | u_{<s})) \right]. \quad (32)$$

Proof of Proposition A.7. Similar to how we showed Proposition A.1, we substitute eq. (30), exchange the two finite sums, and split each inner sum over the prefix $u_{<s}$ and the final token $u_s = w$. The inner sum over w is an entropy and the outer sum is an expectation over the prefixes the producer assigns, hence at every position s we can obtain

$$- \sum_{u \in \mathcal{V}^m} Q(u) \log q(u_s | u_{<s}) = \mathbb{E}_{u_{<s} \sim Q} \left[\mathbb{H}(q(\cdot | u_{<s})) \right].$$

Summing over the m positions yields eq. (32).

The second difference therefore remains, which is the distribution the prefixes $u_{<s}$ of eq. (32) are drawn from.

Proposition A.8 (Unbiased estimate). *If $u \sim Q$, then*

$$\mathbb{E}_{u \sim Q} \left[\widehat{\mathbb{H}}(u) \right] = \frac{1}{m} \mathbb{H}(Y). \quad (33)$$

Proof of Proposition A.8. Marginalizing Q over every continuation past $s-1$,

$$\sum_{u_{\geq s} \in \mathcal{V}^{m-s+1}} Q(u) = Q(u_{<s}),$$

hence $u_{<s} \sim Q$ whenever $u \sim Q$, and

$$\mathbb{E}_{u \sim Q} \left[\mathbb{H}(q(\cdot | u_{<s})) \right] = \mathbb{E}_{u_{<s} \sim Q} \left[\mathbb{H}(q(\cdot | u_{<s})) \right] \quad \text{at every position } s.$$

Averaging these m equalities and applying Proposition A.7,

$$\mathbb{E}_{u \sim Q} \left[\widehat{\mathbb{H}}(u) \right] = \frac{1}{m} \sum_{s=1}^m \mathbb{E}_{u_{<s} \sim Q} \left[\mathbb{H}(q(\cdot | u_{<s})) \right] = \frac{1}{m} \mathbb{H}(Y).$$

A single output supplies one prefix at every position, and Proposition A.8 states that these m prefixes already follow the distribution eq. (32) requires. No enumeration of $\mathcal{V}^m$ is needed, and the second difference reduces to the origin of that single output.

Assumption A.6 (Producer reference faithfulness). Let D_{prod} denote the distribution from which the training data draws the producer’s output u , conditioned on the producer’s prompt. There exists $\varepsilon_{\text{prod}} \in [0, 1]$ with $d_{\text{TV}}(D_{\text{prod}}, Q) \leq \varepsilon_{\text{prod}}$.

Assumption A.6 transposes Assumption A.2 from the consumer to the producer, requiring that the producer, given its own prompt, reproduce what the training data records at its position of the pipeline.

Proposition A.9 (The estimate tracks the axiom). Under Assumption A.6,

$$\left| \mathbb{E}_{u \sim D_{\text{prod}}} [\hat{\mathbb{H}}(u)] - \frac{1}{m} \mathbb{H}(Y) \right| \leq \varepsilon_{\text{prod}} \log |\mathcal{V}|. \quad (34)$$

Proof of Proposition A.9. The entropy of a distribution on $\mathcal{V}$ takes values in $[0, \log |\mathcal{V}|]$. Averaging m of them in eq. (5),

$$0 \leq \hat{\mathbb{H}}(u) = \frac{1}{m} \sum_{s=1}^m \mathbb{H}(q(\cdot | u_{<s})) \leq \frac{1}{m} \sum_{s=1}^m \log |\mathcal{V}| = \log |\mathcal{V}|,$$

gives $\hat{\mathbb{H}}: \mathcal{V}^m \rightarrow [0, \log |\mathcal{V}|]$. Therefore, Lemma A.1 applied to $\hat{\mathbb{H}}$ and to the two distributions D_{prod} and Q on $\mathcal{V}^m$ gives

$$\left| \mathbb{E}_{D_{\text{prod}}} [\hat{\mathbb{H}}] - \mathbb{E}_Q [\hat{\mathbb{H}}] \right| \leq \log |\mathcal{V}| \cdot d_{\text{TV}}(D_{\text{prod}}, Q) \leq \varepsilon_{\text{prod}} \log |\mathcal{V}|,$$

and Proposition A.8 identifies the second expectation as $\mathbb{H}(Y)/m$.

The division by m in eq. (5) renders eq. (34) independent of the output length and makes the target an entropy rate. Without it the target would grow with m , and a long output the producer was certain about would be indistinguishable from a short one it was uncertain about.

Corollary A.4 (The surrogate regresses onto the axiom). Let $\eta(u) = \hat{\mathbb{H}}(u) - \mathbb{H}(Y)/m$, under which eq. (6) takes the form

$$\begin{aligned} \mathcal{L}_{\text{stab}}(\psi, \omega) &= \left(g_{\omega}(\mathbf{T}) - \hat{\mathbb{H}}(u) \right)^2 \\ &= \left(g_{\omega}(\mathbf{T}) - \frac{1}{m} \mathbb{H}(Y) - \eta(u) \right)^2. \end{aligned} \quad (35)$$

Then η is independent of ψ and ω , and under Assumption A.6

$$\mathbb{E}_{u \sim Q} [\eta(u)] = 0, \quad \left| \mathbb{E}_{u \sim D_{\text{prod}}} [\eta(u)] \right| \leq \varepsilon_{\text{prod}} \log |\mathcal{V}|.$$

Consequently every (ψ, ω) at which eq. (6) equals zero recovers $\mathbb{H}(Y)/m$ from $\mathbf{T}$ up to $|\eta(u)|$.

Proof of Corollary A.4. Substituting $\hat{\mathbb{H}}(u) = \mathbb{H}(Y)/m + \eta(u)$ into eq. (6) yields eq. (35). Since η depends only on the frozen producer and on u , it depends on neither ψ nor ω . Subtracting $\mathbb{H}(Y)/m$ from eq. (33) gives

$$\mathbb{E}_{u \sim Q} [\eta(u)] = 0,$$

and the same subtraction inside eq. (34) gives

$$\left| \mathbb{E}_{u \sim D_{\text{prod}}} [\eta(u)] \right| \leq \varepsilon_{\text{prod}} \log |\mathcal{V}|.$$

Finally the right side of eq. (35) equals zero if and only if

$$g_\omega(\mathbf{T}) = \frac{1}{m} \mathbb{H}(Y) + \eta(u).$$

Corollary A.4 shows that a single term accounts for all of the imprecision remaining in eq. (6). That term, $\eta(u)$, comes from the one output that was actually sampled, and it does not depend on ψ or ω . Assumption A.6 keeps its average under the training distribution within $\varepsilon_{\text{prod}} \log |\mathcal{V}|$ of zero. Therefore at $\mathcal{L}_{\text{stab}}(\psi, \omega) = 0$ the probe g_ω recovers the entropy rate $\mathbb{H}(Y)/m$ from $\mathbf{T}$.

A.2 Effect on the Consumer’s Cross-Entropy

This subsection connects each loss term stated in Section 3 to the empirical performance gains observed in Section 4.

Causality

Let CE_{lat} and CE_{txt} denote the per-token cross-entropy that the consumer incurs on the target v under the two transfers of Definition 2.1,

$$\text{CE}_{\text{lat}}(v) = -\frac{1}{n} \log P(v \mid \mathbf{T}), \quad \text{CE}_{\text{txt}}(v) = -\frac{1}{n} \log P(v \mid E(u)), \quad (36)$$

where eq. (8) supplies each at the corresponding block Z . The difference between them is the cost of substituting the thought for the producer’s text,

$$\Lambda(v) \triangleq \text{CE}_{\text{lat}}(v) - \text{CE}_{\text{txt}}(v) = \frac{1}{n} \log \frac{P(v \mid E(u))}{P(v \mid \mathbf{T})}. \quad (37)$$

The average of eq. (37) recovers eq. (9).

Proposition A.10 (The axiom’s divergence is an excess cross-entropy). *For every transfer,*

$$\mathbb{E}_{v \sim P(\cdot \mid E(u))} [\Lambda(v)] = \frac{1}{n} D_{\text{KL}}(P(\cdot \mid E(u)) \parallel P(\cdot \mid \mathbf{T})) \geq 0. \quad (38)$$

Proof of Proposition A.10. Averaging eq. (37) under $P(\cdot \mid E(u))$,

$$\begin{aligned} \mathbb{E}_{v \sim P(\cdot \mid E(u))} [\Lambda(v)] &= \sum_{v \in \mathcal{V}^n} P(v \mid E(u)) \cdot \frac{1}{n} \log \frac{P(v \mid E(u))}{P(v \mid \mathbf{T})} \\ &= \frac{1}{n} \sum_{v \in \mathcal{V}^n} P(v \mid E(u)) \log \frac{P(v \mid E(u))}{P(v \mid \mathbf{T})} \\ &= \frac{1}{n} D_{\text{KL}}(P(\cdot \mid E(u)) \parallel P(\cdot \mid \mathbf{T})) \geq 0, \end{aligned}$$

where Gibbs’ inequality gives the final inequality.

Proposition A.10 averages over the targets that the textual transfer induces, whereas training averages over D . Assumption A.2 bounds the distance between the two distributions and Assumption A.1 bounds the quantity being averaged, hence eq. (2) controls the two cross-entropies of eq. (36) under the conditions that Appendix A.1.1 already states.

Proposition A.11 (The surrogate bounds the excess cross-entropy). *Under Assumptions A.1 and A.2, at every ψ in the optimization domain,*

$$-2B\varepsilon \leq \mathbb{E}_{v \sim D} [\Lambda(v)] \leq \mathbb{E}_{v \sim D} [\mathcal{L}_{\text{caus}}(\psi)] + 3B\varepsilon. \quad (39)$$

Proof of Proposition A.11. Expanding eq. (37) through eq. (8) and bounding each of its n log-ratios by eq. (13),

$$|\Lambda(v)| = \left| \frac{1}{n} \sum_{t=1}^n \log \frac{p(v_t \mid v_{<t}, E(u))}{p(v_t \mid v_{<t}, \mathbf{T})} \right| \leq \frac{1}{n} \sum_{t=1}^n B = B,$$

hence $\Lambda : \mathcal{V}^n \rightarrow [-B, B]$ and its range fits in an interval of length $2B$. Lemma A.1, applied to Λ and to the two distributions D and $P(\cdot | E(u))$ on $\mathcal{V}^n$, therefore gives

$$\left| \mathbb{E}_D[\Lambda] - \mathbb{E}_{P(\cdot | E(u))}[\Lambda] \right| \leq 2B \cdot d_{\text{TV}}(D, P(\cdot | E(u))) \leq 2B\varepsilon. \quad (40)$$

Combining eq. (40) with eq. (38) on the left and with Proposition A.2 on the right yields

$$\begin{aligned} \mathbb{E}_D[\Lambda] &\geq \mathbb{E}_{P(\cdot | E(u))}[\Lambda] - 2B\varepsilon \geq -2B\varepsilon, \\ \mathbb{E}_D[\Lambda] &\leq \frac{1}{n} D_{\text{KL}}(P(\cdot | E(u)) \| P(\cdot | \mathbf{T})) + 2B\varepsilon \leq \mathbb{E}_{v \sim D}[\mathcal{L}_{\text{caus}}(\psi)] + 3B\varepsilon. \end{aligned}$$

Corollary A.5 (Parity at zero surrogate). *If $\mathcal{L}_{\text{caus}}(\psi) = 0$, then $\Lambda(v) = 0$.*

Proof of Corollary A.5. Equation (2) averages n nonnegative divergences, hence

$$\mathcal{L}_{\text{caus}}(\psi) = 0 \quad \implies \quad D_{\text{KL}}(p(\cdot | v_{<t}, E(u)) \| p(\cdot | v_{<t}, \mathbf{T})) = 0 \quad \text{at every } t,$$

and a divergence vanishes only at equal distributions,

$$p(\cdot | v_{<t}, E(u)) = p(\cdot | v_{<t}, \mathbf{T}) \quad \text{at every } t.$$

Substituting into eq. (8),

$$P(v | E(u)) = \prod_{t=1}^n p(v_t | v_{<t}, E(u)) = \prod_{t=1}^n p(v_t | v_{<t}, \mathbf{T}) = P(v | \mathbf{T}),$$

at which eq. (37) gives $\Lambda(v) = 0$.

Corollary A.5 holds at a single v and therefore requires neither assumption. Exponentiating eq. (37) restates eq. (39) in terms of the probability that the consumer assigns to the correct answer.

Corollary A.6 (Discount on the correct answer). *For every v ,*

$$P(v | \mathbf{T}) = P(v | E(u)) \cdot \exp(-n\Lambda(v)), \quad (41)$$

and under Assumptions A.1 and A.2 the geometric mean of the discount factor under D satisfies

$$\exp\left(-n(\mathbb{E}_{v \sim D}[\mathcal{L}_{\text{caus}}(\psi)] + 3B\varepsilon)\right) \leq \exp\left(-n\mathbb{E}_{v \sim D}[\Lambda(v)]\right) \leq \exp(2nB\varepsilon). \quad (42)$$

Proof of Corollary A.6. Multiplying eq. (37) by n and exponentiating,

$$\exp(n\Lambda(v)) = \frac{P(v | E(u))}{P(v | \mathbf{T})},$$

and rearranging yields eq. (41). The geometric mean of its discount factor under D is

$$\exp\left(\mathbb{E}_{v \sim D}[\log \exp(-n\Lambda(v))]\right) = \exp\left(-n\mathbb{E}_{v \sim D}[\Lambda(v)]\right),$$

and the exponential is increasing, hence applying it to the two bounds of eq. (39) yields eq. (42).

Within eq. (39), the data and the frozen consumer fix every quantity except $\mathbb{E}_D[\mathcal{L}_{\text{caus}}]$. Equation (2) therefore determines the substitution cost, and Corollary A.5 establishes that reducing it to zero recovers the textual transfer exactly.

Minimality

Let V denote the random variable realized by the consumer's target v of Definition 2.1, and condition every entropy below on the blocks E_{pre} and E_{post} that Definition 2.1 holds independent of ψ , left implicit in the notation. The smallest cross-entropy any decoder incurs on V under a transferred block is the conditional entropy of V given that block, hence

$$\mathbb{H}(V | \mathbf{T}) - \mathbb{H}(V | Y) \quad (43)$$

is the cost of substituting the thought for the producer's output. We establish that eq. (3) upper-bounds eq. (43) up to a constant, hence lowering the loss lowers an upper bound on that cost.

Assumption A.7 (Nested input probes). Every probe that predicts X from Y is available to the probe that predicts X from $(Y, \mathbf{T})$, which recovers it by discarding $\mathbf{T}$.

Comparing the two probe classes at their optima² under Assumption A.7 bounds one cross-entropy of eq. (3),

$$\text{CE}(X | Y, \mathbf{T}) \leq \text{CE}(X | Y), \quad (44)$$

and their difference estimates the residual mutual information $I(X; \mathbf{T} | Y)$,

$$\hat{I}(X; \mathbf{T} | Y) \triangleq \text{CE}(X | Y) - \text{CE}(X | Y, \mathbf{T}) \geq 0. \quad (45)$$

Only $\text{CE}(X | Y, \mathbf{T})$ of eq. (45) is in terms of $\mathbf{T}$, hence $\text{CE}(X | Y)$ is constant in ψ .

Proposition A.12 (The loss bounds the reconstruction cross-entropy). Under Assumption A.7, at every ψ in the optimization domain,

$$\text{CE}(Y | \mathbf{T}) \leq \frac{1}{\lambda_1} \left(\mathcal{L}_{\min}(\psi) + \lambda_2 \text{CE}(X | Y) \right), \quad (46)$$

and the difference between the two sides of eq. (46) is

$$\frac{1}{\lambda_1} \left(\mathcal{L}_{\min}(\psi) + \lambda_2 \text{CE}(X | Y) \right) - \text{CE}(Y | \mathbf{T}) = \frac{\lambda_2}{\lambda_1} \hat{I}(X; \mathbf{T} | Y). \quad (47)$$

Proof of Proposition A.12. Solving eq. (3) for its first term, which $\lambda_1 > 0$ permits,

$$\text{CE}(Y | \mathbf{T}) = \frac{1}{\lambda_1} \left(\mathcal{L}_{\min}(\psi) + \lambda_2 \text{CE}(X | Y, \mathbf{T}) \right).$$

The remaining cross-entropy appears with the positive coefficient λ_2/λ_1 , hence replacing it by the larger quantity of eq. (44) preserves the inequality and yields eq. (46). Subtracting $\text{CE}(Y | \mathbf{T})$ from the right side of eq. (46) and applying the same identity,

$$\frac{1}{\lambda_1} \left(\mathcal{L}_{\min}(\psi) + \lambda_2 \text{CE}(X | Y) \right) - \text{CE}(Y | \mathbf{T}) = \frac{\lambda_2}{\lambda_1} \left(\text{CE}(X | Y) - \text{CE}(X | Y, \mathbf{T}) \right),$$

which is eq. (47) by eq. (45).

Proposition A.13 (Reconstruction uncertainty bounds answer uncertainty). For every transfer,

$$\mathbb{H}(V | \mathbf{T}) - \mathbb{H}(V | Y) \leq \mathbb{H}(Y | \mathbf{T}). \quad (48)$$

²Assumption A.7 also requires $\text{CE}(X | Y, \mathbf{T}) \leq \text{CE}(X | Y)$ for the frozen consumer, that is, adding the thought never makes it worse at reconstructing X .

Proof of Proposition A.13. Adjoining a variable cannot lower a joint entropy, the chain rule splits the result, and conditioning on more cannot raise an entropy,

$$\begin{aligned}\mathbb{H}(V | \mathbf{T}) &\leq \mathbb{H}(V, Y | \mathbf{T}) \\ &= \mathbb{H}(Y | \mathbf{T}) + \mathbb{H}(V | Y, \mathbf{T}) \\ &\leq \mathbb{H}(Y | \mathbf{T}) + \mathbb{H}(V | Y),\end{aligned}$$

and subtracting $\mathbb{H}(V | Y)$ yields eq. (48).

Proposition A.14 (The surrogate bounds the substitution cost). *Under Assumption A.7, at every ψ in the optimization domain,*

$$\mathbb{H}(V | \mathbf{T}) - \mathbb{H}(V | Y) \leq \frac{1}{\lambda_1} \left(\mathcal{L}_{\min}(\psi) + \lambda_2 \text{CE}(X | Y) \right). \quad (49)$$

Proof of Proposition A.14. Gibbs' inequality bounds an entropy by the cross-entropy that approximates it,

$$\mathbb{H}(Y | \mathbf{T}) \leq \text{CE}(Y | \mathbf{T}),$$

and chaining eq. (48), this inequality, and eq. (46) gives

$$\mathbb{H}(V | \mathbf{T}) - \mathbb{H}(V | Y) \leq \mathbb{H}(Y | \mathbf{T}) \leq \text{CE}(Y | \mathbf{T}) \leq \frac{1}{\lambda_1} \left(\mathcal{L}_{\min}(\psi) + \lambda_2 \text{CE}(X | Y) \right).$$

The data and the frozen models fix $\text{CE}(X | Y)$ and $\mathbb{H}(V | Y)$. We can restate eq. (49) in terms of the probability that the consumer assigns to its target.

Corollary A.7 (Bound on the probability of the target). *Under Assumption A.7, the geometric mean probability that an optimal decoder assigns to the consumer's target satisfies*

$$\exp \left(-\mathbb{H}(V | \mathbf{T}) \right) \geq \exp \left(-\mathbb{H}(V | Y) \right) \cdot \exp \left(-\frac{1}{\lambda_1} \left(\mathcal{L}_{\min}(\psi) + \lambda_2 \text{CE}(X | Y) \right) \right). \quad (50)$$

Proof of Corollary A.7. The exponential is increasing, hence negating eq. (49) and applying it gives

$$\exp \left(-\mathbb{H}(V | \mathbf{T}) \right) \geq \exp \left(-\mathbb{H}(V | Y) - \frac{1}{\lambda_1} \left(\mathcal{L}_{\min}(\psi) + \lambda_2 \text{CE}(X | Y) \right) \right),$$

and splitting the exponential of a sum yields eq. (50). The left side is the reciprocal perplexity of V under $\mathbf{T}$, which is the geometric mean of the probability assigned to the target.

The exponential of eq. (50) does not exceed one, and it attains one exactly when $\text{CE}(Y | \mathbf{T})$ and $\hat{I}(X; \mathbf{T} | Y)$ both vanish, at which the minimality loss reaches its minimum $-\lambda_2 \text{CE}(X | Y)$ and the probability under the latent transfer attains the probability under the textual one.

Separability

Let v denote the consumer's target under $\mathbf{T}$ and let v_k denote the target of the k -th preceding example under $\mathbf{T}_k$, with

$$a = P(v | \mathbf{T}), \quad a_k = P(v_k | \mathbf{T}_k), \quad (51)$$

supplied by eq. (8) at the corresponding block.

Proposition A.15 (The loss bounds the nearest distance). *Let k^* attain $\max_k \text{sim}(\mathbf{t}, \mathbf{t}_k)$ and let*

$$\delta \triangleq \sqrt{2 - 2\tau(\mathcal{L}_{\text{sep}}(\psi, \omega) - \log K)}. \quad (52)$$

Then δ is real and

$$\|\varphi_\omega(\mathbf{T}) - \varphi_\omega(\mathbf{T}_{k^*})\| \leq \delta. \quad (53)$$

Proof of Proposition A.15. Bounding each term of eq. (4) by the largest,

$$\begin{aligned}\mathcal{L}_{\text{sep}}(\psi, \omega) &= \log \sum_{k=1}^K \exp(\text{sim}(\mathbf{t}, \mathbf{t}_k) / \tau) \\ &\leq \log \left(K \exp(\text{sim}(\mathbf{t}, \mathbf{t}_{k^*}) / \tau) \right) \\ &= \log K + \frac{1}{\tau} \text{sim}(\mathbf{t}, \mathbf{t}_{k^*}),\end{aligned}\tag{54}$$

and rearranging eq. (54),

$$\text{sim}(\mathbf{t}, \mathbf{t}_{k^*}) \geq \tau (\mathcal{L}_{\text{sep}}(\psi, \omega) - \log K).\tag{55}$$

Cosine similarity does not exceed one, hence eq. (55) lower-bounds the radicand of eq. (52) by zero and δ is real. Substituting eq. (55) into eq. (26),

$$\begin{aligned}\|\varphi_\omega(\mathbf{T}) - \varphi_\omega(\mathbf{T}_{k^*})\|^2 &= 2 - 2\text{sim}(\mathbf{t}, \mathbf{t}_{k^*}) \\ &\leq 2 - 2\tau (\mathcal{L}_{\text{sep}}(\psi, \omega) - \log K) = \delta^2,\end{aligned}$$

and taking square roots yields eq. (53).

Assumption A.8 (Smooth consumer). There exists a finite L with

$$d_{\text{TV}}(P(\cdot | \mathbf{T}), P(\cdot | \mathbf{T}')) \leq L \|\varphi_\omega(\mathbf{T}) - \varphi_\omega(\mathbf{T}')\|\tag{56}$$

at every pair of thoughts and every ψ in the optimization domain.

Assumption A.8 requires that the consumer's target distribution vary continuously with the thought it receives, and it identifies that distribution's dependence on $\mathbf{T}$ with the pooled direction $\varphi_\omega(\mathbf{T})$ of eq. (25).

Proposition A.16 (Colliding thoughts share one budget). Under Assumptions A.5 and A.8,

$$a + a_{k^*} \leq 1 + L\delta.\tag{57}$$

Proof of Proposition A.16. Applying eq. (56) to the single event $\{v\}$ and then eq. (53),

$$P(v | \mathbf{T}_{k^*}) \geq P(v | \mathbf{T}) - L \|\varphi_\omega(\mathbf{T}) - \varphi_\omega(\mathbf{T}_{k^*})\| \geq a - L\delta.\tag{58}$$

Assumption A.5 makes the two examples semantically disjoint, hence v and v_{k^*} are distinct outcomes of the single distribution $P(\cdot | \mathbf{T}_{k^*})$ and

$$P(v | \mathbf{T}_{k^*}) + P(v_{k^*} | \mathbf{T}_{k^*}) \leq 1.\tag{59}$$

Substituting eq. (58) into eq. (59),

$$(a - L\delta) + a_{k^*} \leq P(v | \mathbf{T}_{k^*}) + P(v_{k^*} | \mathbf{T}_{k^*}) \leq 1,$$

and adding $L\delta$ to both ends yields eq. (57).

Proposition A.17 (The loss floors the cross-entropy). Under Assumptions A.5 and A.8, at every ψ in the optimization domain,

$$\frac{1}{2}(-\log a - \log a_{k^*}) \geq \log 2 - \log(1 + L\delta).\tag{60}$$

Proof of Proposition A.17. The inequality of arithmetic and geometric means bounds the product by the square of the mean, and eq. (57) bounds that mean,

$$a a_{k^*} \leq \left(\frac{a + a_{k^*}}{2} \right)^2 \leq \left(\frac{1 + L\delta}{2} \right)^2. \quad (61)$$

The logarithm is increasing, hence applying $-\frac{1}{2}\log(\cdot)$ to eq. (61) reverses it,

$$\begin{aligned} \frac{1}{2} \left(-\log a - \log a_{k^*} \right) &= -\frac{1}{2} \log(a a_{k^*}) \\ &\geq -\frac{1}{2} \log \left(\frac{1 + L\delta}{2} \right)^2 = -\log \left(\frac{1 + L\delta}{2} \right) = \log 2 - \log(1 + L\delta). \end{aligned}$$

Substituting eq. (52) states eq. (60) through the loss,

$$\frac{1}{2} \left(-\log a - \log a_{k^*} \right) \geq \log 2 - \log \left(1 + L \sqrt{2 - 2\tau(\mathcal{L}_{\text{sep}}(\psi, \omega) - \log K)} \right). \quad (62)$$

Corollary A.8 (Threshold above which the objective is obstructed). *Under Assumptions A.5 and A.8, the right side of eq. (62) is positive if and only if*

$$\mathcal{L}_{\text{sep}}(\psi, \omega) > \log K + \frac{1}{\tau} \left(1 - \frac{1}{2L^2} \right). \quad (63)$$

Proof of Corollary A.8. Positivity of the right side of eq. (60) requires

$$\log 2 > \log(1 + L\delta) \quad \Longleftrightarrow \quad L\delta < 1 \quad \Longleftrightarrow \quad \delta^2 < \frac{1}{L^2},$$

and substituting eq. (52),

$$\begin{aligned} 2 - 2\tau(\mathcal{L}_{\text{sep}}(\psi, \omega) - \log K) &< \frac{1}{L^2} \\ \mathcal{L}_{\text{sep}}(\psi, \omega) &> \log K + \frac{1}{\tau} \left(1 - \frac{1}{2L^2} \right). \end{aligned}$$

If training resulted in two different examples having colliding thoughts, then a consumer agent must then answer both with nearly the same distribution even if the two target texts are different. This imposes a lower-bound on their average cross-entropy (eq. (62)). Under total collapse, where $\delta = 0$, the optimal solution would be assigning equal probability to both targets which will make the lower-bound at its largest value ($\log 2$, random chance between two targets). CE loss does not prevent such collisions through its objective.

Stability

A textual transfer delivers one output u drawn from the producer's output distribution Q of eq. (30), whereas $\mathbf{T}$ is a trained function and is not bound to that sample. Let $s(u)$ denote the probability the consumer assigns to its target v under the transfer of u , and let $\bar{P}$ denote the mixture over the producer's samples,

$$s(u) = P(v \mid E(u)) \in (0, 1], \quad \bar{s} = \mathbb{E}_{u \sim Q}[s(u)], \quad \bar{P} = \mathbb{E}_{u \sim Q}[P(\cdot \mid E(u))]. \quad (64)$$

Evaluating $\bar{P}$ at v ,

$$\bar{P}(v) = \mathbb{E}_{u \sim Q}[P(v \mid E(u))] = \bar{s}. \quad (65)$$

Both proofs below use

$$f(x) = -\log x, \quad f''(x) = \frac{1}{x^2} \geq 1 \quad \text{on } (0, 1]. \quad (66)$$

Proposition A.18 (The mixture bounds the sampled transfer). *For every producer,*

$$G \triangleq \mathbb{E}_{u \sim Q}[-\log s(u)] - (-\log \bar{s}) \geq 0, \quad (67)$$

with equality if and only if

$$s(u) = \bar{s} \quad \text{for } Q\text{-almost every } u. \quad (68)$$

Proof of Proposition A.18. Equation (66) makes f strictly convex on $(0, 1]$, hence Jensen's inequality gives

$$\mathbb{E}_{u \sim Q}[-\log s(u)] \geq -\log \mathbb{E}_{u \sim Q}[s(u)] = -\log \bar{s},$$

and strict convexity attains equality only at eq. (68).

Proposition A.19 (The advantage is the sampling variance). *For every producer,*

$$G \geq \frac{1}{2} \text{Var}_{u \sim Q}(s(u)). \quad (69)$$

Proof of Proposition A.19. Taylor's theorem at $\bar{s}$ with Lagrange remainder places ξ between $s(u)$ and $\bar{s}$, hence in $(0, 1]$, and eq. (66) bounds $f''(\xi)$,

$$\begin{aligned} -\log s(u) &= -\log \bar{s} - \frac{s(u) - \bar{s}}{\bar{s}} + \frac{1}{2} f''(\xi) (s(u) - \bar{s})^2 \\ &\geq -\log \bar{s} - \frac{s(u) - \bar{s}}{\bar{s}} + \frac{1}{2} (s(u) - \bar{s})^2. \end{aligned}$$

The linear term vanishes under Q ,

$$\mathbb{E}_{u \sim Q}[s(u) - \bar{s}] = 0,$$

hence averaging the previous display over $u \sim Q$,

$$\mathbb{E}_{u \sim Q}[-\log s(u)] \geq -\log \bar{s} + \frac{1}{2} \mathbb{E}_{u \sim Q}[(s(u) - \bar{s})^2] = -\log \bar{s} + \frac{1}{2} \text{Var}_{u \sim Q}(s(u)),$$

and subtracting $-\log \bar{s}$ yields eq. (69).

Assumption A.9 (Realizable mixture). Some ψ in the optimization domain satisfies

$$P(\cdot | \mathbf{T}) = \bar{P}. \quad (70)$$

Corollary A.9 (The latent transfer attains the advantage). *Under Assumption A.9, at that ψ ,*

$$-\log P(v | \mathbf{T}) = \mathbb{E}_{u \sim Q}[-\log P(v | E(u))] - G \leq \mathbb{E}_{u \sim Q}[-\log P(v | E(u))]. \quad (71)$$

Proof of Corollary A.9. Evaluating eq. (70) at the single outcome v and applying eq. (65),

$$-\log P(v | \mathbf{T}) = -\log \bar{P}(v) = -\log \bar{s},$$

and eq. (67) rearranges this to

$$-\log P(v | \mathbf{T}) = \mathbb{E}_{u \sim Q}[-\log s(u)] - G,$$

where $G \geq 0$ by Proposition A.18.

A textual transfer cannot attain the left side of eq. (71), since $E(u)$ is the sampled output and its target distribution is fixed at $P(\cdot | E(u))$. Equation (69) makes the advantage strictly positive whenever the producer's sample moves the consumer's target probability. Equation (6) encodes a scalar summary of Q , since encoding Q requires multiple producer samples per training example.

B Implementation Details

B.1 Benchmarks

- **MATH500** [29] contains 500 test problems drawn uniformly at random from MATH, and its mix of subjects and difficulty levels is representative of the full test set.
- **AIME2025** [52] is the 2025 American Invitational Mathematics Examination, 30 problems whose answers are single integers.
- **AIME2026** [12] is the 2026 examination in the same format.
- **GPQA-Diamond** [36] is the most strictly filtered GPQA subset, 198 graduate-level multiple-choice questions in biology, chemistry and physics that both expert validators answered correctly and most non-experts missed.
- **MedQA** [23] draws multiple-choice questions from medical licensing examinations.
- **LiveCodeBench-v6** [22] collects competition programming problems with release dates, which lets evaluation be restricted to a window after a model’s training cutoff.
- **MBPP+** [30] re-evaluates the Python programming tasks of MBPP against many more test cases, and a program counts as correct only when it passes every test.

We report Pass@10 for AIME2025 and AIME2026, and Pass@1 for every other benchmark. The **Code Gen.** column reports MBPP+ on the Light system and LiveCodeBench-v6 on the Scaled system.

B.2 Compared Baselines

We compare REST against the following alternative auxiliary-loss baselines:

- **CODI** [40] trains continuous latent thoughts as the student in a self-distillation setup, matching the student’s hidden states at one distillation token to those of a teacher pass that reads the explicit chain of thought. We adapt its distillation loss as an auxiliary term on the outer-link cross-entropy objective, taking the consumer’s text transfer of Definition 2.1 as the teacher.
- **SIM-CoT** [46] supervises latent reasoning with an auxiliary decoder that reconstructs reasoning-step text from the implicit latents and is discarded at inference. We adapt its reconstruction loss as an auxiliary term on the outer-link cross-entropy objective, training a per-stage decoder to reconstruct the producer’s output text from the transferred thought.

B.3 Hyperparameters

Optimization. We train the outer links for 20,000 steps at batch size 4 and a maximum sequence length of 4096, in bfloat16. AdamW runs at a learning rate of 5×10^{-4} under a cosine schedule with 10 warmup steps, no weight decay, and gradients clipped at 1.0.

Loss Terms. Every property term is swept over $\beta \in \{0.1, 0.3, 1.0, 3.0\}$. For minimality, we set $\lambda_1 = 1.0$ and $\lambda_2 = 0.3$ and 0 for composition. For separability, we set $K = 64$ at $\tau = 0.1$. The CODI baseline is swept over $\{1, 10, 20\}$ and SIM-CoT over $\{0.3, 1, 3\}$.

Inference. A thought spans at most 80 positions in training and a fixed 32 latent steps at inference. We sample at temperature 0.6 and top- p 0.95, lowered to 0.2 on MBPP+ and LiveCodeBench-v6. Table 5 provides max length for generated text.

Table 5. Generation budget in tokens.

Benchmark	Light	Scaled
MATH500	1000	2000
AIME2025	8192	16000
AIME2026	8192	16000
GPQA-D	4000	4000
MedQA	4000	4000
MBPP+	4000	4000
LCB-v6	4096	4096

C Extended Related Work

Reasoning in Latent Space. A parallel family internalizes explicit rationales into the weights or the architecture, leaving the input stream unchanged. Stepwise removal of thinking tokens moves an explicit chain of thought into the forward pass [14], dense embeddings stand in for a compressed chain [8], and latent and text tokens are mixed within one sequence [42]. Architectural variants add depth or recurrence at fixed points of the network, through dynamic depth scaling [7] and through

middle-layer recurrence across decoding steps [3]. Recurrence can also carry a persistent latent state across refinement steps, each supervised against a ground-truth intermediate target [27]. Huang et al. [21] prove that a transformer internalizes a chain of thought under a staged curriculum, where the schedule for removing thinking tokens sets the number of training stages. Latent reasoning has also been applied to chemistry [51], built on hybrid state-space backbones [44], and cast as an exchange between a fast and a slow process [9].

Supervising Latent Thoughts. Two objectives compress reasoning over decoded traces rather than over latent representations. Massoli et al. [32] model a reasoning trace under a conditional information bottleneck, retaining only the information about the response that the prompt does not already supply, and optimize it as a reinforcement learning objective with a surprisal prior over traces. Conklin et al. [10] treat training itself as lossy compression and report that pretrained models approach the information bottleneck bound, which links how much a model compresses to downstream performance. Neither objective constrains a thought transferred between agents. REST applies its terms at that transfer. Other work measures internal states without training on them, through uncertainty over the meanings a model could generate [25] and through the geometry of reasoning trajectories in representation space [55]. Li et al. [28] intervene on individual latent steps and find that the steps carry different functions and route information non-locally. A larger latent budget does not act as uniform extra depth.

REST differs in the object it constrains and in the form of the constraint. Its terms apply to the latent thought at the point where one agent or one round passes it to the next, and each term is a differentiable addition to the cross-entropy objective that leaves the architecture unchanged and adds no parameters at inference.

D Additional Results

D.1 Base Model Accuracy

As shown in Table 6, each frozen model answers at a level that varies substantially across domains, and no single model leads on every benchmark within either agent composition. The latent system combines agents of complementary strength, and the values here give the accuracy of each agent before any system is applied.

Table 6. Frozen base LLMs, no system.

Model	Metric	MATH500	GPQA-D	MedQA	AIME25	AIME26	MBPP+	LCB-v6
Qwen3-1.7B	Acc.	68.4	33.5	46.2	20.0	22.2	57.3	22.8
	Token	546	1009	472	1962	2695	80	608
Llama-3.2-1B	Acc.	25.0	27.4	37.0	1.1	2.2	41.7	4.3
	Token	417	629	369	2412	2225	309	288
Qwen2.5-Math-1.5B	Acc.	76.3	29.6	28.4	27.8	22.2	33.1	2.9
	Token	530	807	1061	908	987	693	669
Gemma-3-4B	Acc.	76.3	31.0	52.9	27.8	23.3	70.8	18.5
	Token	829	807	341	1543	1623	136	278
Llama-3.2-3B	Acc.	49.6	31.0	58.4	10.0	7.8	59.5	12.0
	Token	509	967	385	3389	3033	287	324
Qwen3.5-4B	Acc.	79.1	62.5	83.8	81.1	90.0	70.9	41.1
	Token	993	2526	1086	10752	10888	272	1883

D.2 Per-Round Results

Table 2 and Table 3 in the main text configure each system at its own recursion round, and compare every method against the CE-only baseline at that same round. The tables below give both rounds in full for each setting. Across training seeds, the average standard error of the Avg. Change column is ± 0.7 points in accuracy and $\pm 2.1\%$ in tokens.

As shown in Tables 7 to 10 and summarised in Figure 4, recursion depth affects the two scales in opposite directions. In the Light setting, every property term improves over the CE-only objective at $r = 1$, and the improvement is diminished at $r = 3$. In the Scaled setting the ordering is reversed. REST is relatively close to CE-only at $r = 1$, while at $r = 3$ every term except multi-agent separability improves accuracy. At $r = 1$ our CE-only reproduction is close to the accuracy reported by Zou et al. [58] on the Scaled system and lower on the Light system, while at $r = 3$ it is lower on both, most of all on AIME, where seed variance is largest. Avg. Change at $r = 3$ is measured against this reproduction of the experiment.

Table 7. Single-agent (round $r = 1$), Light vs Scaled.

Method	Metric	Math500		AIME2025		AIME2026		GPQA-D		MedQA		Code Gen.		Avg. Change	
		Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled
CE only	Acc.	70.6	81.1	27.8	80.0	12.2	80.0	27.3	61.8	27.1	79.8	27.7	37.3	Base	Base
	Token	557	880	905	8272	1016	8468	911	2081	1177	767	477	1353	Base	Base
REST (ours), SINGLE PROPERTY															
Causality ($\beta = 0.3$)	Acc.	72.4	83.3	23.3	68.9	16.7	76.7	23.9	62.8	29.7	79.9	32.9	38.1	$\uparrow 1.0$	$\downarrow 1.7$
	Token	551	842	907	8335	1008	7292	851	1999	1113	739	570	1369	-0.9%	-5.7%
Minimality ($\beta = 1.0$)	Acc.	72.1	79.3	30.0	77.8	20.0	83.3	27.6	59.8	29.7	80.8	31.5	40.9	$\uparrow 3.0$	$\uparrow 0.3$
	Token	546	1000	906	9839	936	9602	872	2532	1075	1108	561	1797	-2.9%	$+18.6\%$
Separability ($\beta = 0.1$)	Acc.	73.1	79.9	26.7	81.1	17.8	83.3	28.8	64.8	27.7	78.8	30.2	41.3	$\uparrow 1.9$	$\uparrow 1.5$
	Token	551	974	918	9324	987	8872	901	2408	1083	904	488	1799	-2.3%	$+11.3\%$
Stability ($\beta = 1.0$)	Acc.	72.1	79.9	27.8	72.2	15.6	77.8	28.1	63.5	29.8	80.1	29.1	37.8	$\uparrow 1.6$	$\downarrow 1.5$
	Token	538	894	921	8334	944	7250	845	2197	1051	804	645	1367	-2.0%	-4.5%
REST (ours), COMPOSITION OF PROPERTIES															
Best pair	Acc.	69.7	80.9	23.3	81.1	16.7	81.1	26.1	59.9	29.6	82.0	33.5	40.9	$\uparrow 1.0$	$\uparrow 1.0$
	Token	542	951	876	9160	1039	8758	826	2433	1083	1013	553	1780	-2.4%	$+10.4\%$
All properties	Acc.	72.4	82.2	33.3	73.3	16.7	83.3	23.7	59.6	31.0	81.7	32.8	39.1	$\uparrow 2.9$	$\downarrow 0.1$
	Token	569	923	934	8749	984	8642	953	2210	1130	833	555	1628	$+1.6\%$	$+5.3\%$

Table 8. Single-agent (round $r = 3$), Light vs Scaled.

Method	Metric	Math500		AIME2025		AIME2026		GPQA-D		MedQA		Code Gen.		Avg. Change	
		Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled
CE only	Acc.	70.9	80.9	26.7	65.6	20.0	72.2	26.9	63.3	27.8	79.2	28.1	35.5	Base	Base
	Token	613	862	912	7226	1020	6967	941	1709	1312	739	667	1289	Base	Base
REST (ours), SINGLE PROPERTY															
Causality ($\beta = 0.3$)	Acc.	74.9	80.4	26.7	78.9	18.9	83.3	29.1	63.5	27.6	79.4	30.0	40.2	$\uparrow 1.1$	$\uparrow 4.8$
	Token	619	1008	910	8677	952	8165	899	2216	1112	955	654	1662	-5.8%	$+20.7\%$
Minimality ($\beta = 1.0$)	Acc.	72.1	81.1	24.4	78.9	16.7	86.7	26.1	63.8	29.2	83.0	32.8	42.3	$\uparrow 0.1$	$\uparrow 6.5$
	Token	627	1053	939	10256	976	10390	888	2526	1236	1162	681	1850	-2.1%	$+44.9\%$
Separability ($\beta = 0.1$)	Acc.	62.0	80.2	26.7	76.7	20.0	83.3	30.8	61.1	31.0	81.7	24.9	39.4	$\downarrow 0.8$	$\uparrow 4.3$
	Token	589	1056	1160	8657	1631	8033	1204	2556	1362	1087	1387	1764	$+34.2\%$	$+23.2\%$
Stability ($\beta = 1.0$)	Acc.	73.3	79.1	26.7	83.3	12.2	88.3	24.7	64.4	31.4	79.8	34.4	40.4	$\uparrow 0.4$	$\uparrow 6.4$
	Token	594	1049	902	9626	1076	9174	865	2464	1043	1071	616	1783	-6.7%	$+33.9\%$
REST (ours), COMPOSITION OF PROPERTIES															
Best pair	Acc.	72.5	79.4	26.7	80.0	18.9	86.7	25.4	57.6	30.2	79.0	31.8	38.6	$\uparrow 0.8$	$\uparrow 4.1$
	Token	618	1065	901	10503	1010	10728	850	2605	995	1179	656	1632	-7.9%	$+47.5\%$
All properties	Acc.	74.9	78.4	26.7	73.3	18.9	76.7	26.6	60.6	30.0	79.3	25.7	37.0	$\uparrow 0.4$	$\uparrow 1.4$
	Token	611	981	925	11053	1030	10338	964	1744	1114	1001	671	1375	-2.7%	$+41.0\%$

D.3 Full Property Sweep

As shown in Table 11, the four property terms separate into two groups. Causality and minimality improve over the CE-only objective at every weight in both settings, indicating that neither term depends on a tuned value of β . Separability and stability remain close to the CE-only objective in the Light setting and fall below it in the Scaled setting at every weight, and the loss of accuracy is accompanied by a substantial reduction in decoded tokens.

Table 9. Multi-agent (round $r = 1$), Light vs Scaled.

Method	Metric	Math500		AIME2025		AIME2026		GPQA-D		MedQA		Code Gen.		Avg. Change	
		Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled
CE only	Acc.	71.1	87.7	22.2	76.7	17.8	86.7	24.9	63.1	28.9	80.2	32.5	39.0	Base	Base
	Token	550	1018	849	9009	884	8488	749	2379	868	1137	604	1507	Base	Base
REST (ours), SINGLE PROPERTY															
Causality	Acc.	77.8	86.4	27.8	80.0	22.2	85.6	28.6	63.1	30.0	82.7	33.9	39.5	↑3.8	↑0.7
	Token	601	1041	894	9538	978	9430	888	2285	1049	1115	643	1545	+12.2%	+6.0%
Minimality	Acc.	77.1	88.0	30.0	83.3	22.2	90.0	27.6	67.2	26.8	81.3	32.0	42.3	↑3.1	↑3.1
	Token	605	1044	920	10309	1052	9956	849	2287	1158	1120	575	2036	+14.6%	+13.7%
Separability	Acc.	74.7	85.6	25.6	56.7	17.8	60.0	25.6	53.5	30.9	77.7	31.1	30.0	↑1.4	↓11.6
	Token	564	790	838	6648	923	6180	758	1084	1116	582	1650	744	+29.9%	-31.9%
Stability	Acc.	74.0	79.0	27.8	40.0	16.7	46.7	26.8	52.0	31.0	77.3	35.1	31.6	↑2.3	↓17.8
	Token	551	581	881	5220	976	5369	785	836	1205	445	781	882	+15.0%	-43.4%
REST (ours), COMPOSITION OF PROPERTIES															
Best pair	Acc.	77.6	87.2	25.6	83.3	15.6	86.7	29.0	67.2	30.1	84.3	33.9	40.0	↑2.4	↑2.6
	Token	604	1018	916	10111	978	7694	886	2206	1100	1103	613	1495	+13.2%	+0.4%
All properties	Acc.	76.2	86.6	28.9	73.3	21.1	86.7	27.6	63.6	28.7	80.0	33.3	33.7	↑3.1	↓1.6
	Token	608	832	900	7634	1022	7466	879	2406	1042	1024	569	962	+11.5%	-13.7%

Table 10. Multi-agent (round $r = 3$), Light vs Scaled.

Method	Metric	Math500		AIME2025		AIME2026		GPQA-D		MedQA		Code Gen.		Avg. Change	
		Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled
CE only	Acc.	70.9	86.6	22.2	80.0	15.6	60.0	27.3	59.1	28.9	79.7	30.3	33.5	Base	Base
	Token	716	1252	796	10485	891	5420	973	2572	1145	829	857	1029	Base	Base
REST (ours), SINGLE PROPERTY															
Causality	Acc.	75.4	86.8	30.0	80.0	14.4	86.7	28.3	54.9	28.1	78.7	34.1	39.1	↑2.5	↑4.6
	Token	792	1158	933	8576	1024	8278	1087	1568	1304	996	777	1924	+10.0%	+4.2%
Minimality	Acc.	67.2	85.8	33.3	86.7	15.0	83.3	27.3	62.6	28.3	83.0	30.4	39.0	↑1.0	↑6.9
	Token	800	1204	1072	9873	1148	9558	1322	2528	1440	1275	889	1875	+24.0%	+21.9%
Separability	Acc.	75.8	82.2	25.6	73.3	16.7	65.6	27.4	52.5	25.4	78.7	27.9	31.6	↑0.6	↓2.5
	Token	753	1191	885	7039	980	6940	1086	1940	1374	1189	867	1280	+10.6%	-9.3%
Stability	Acc.	72.1	85.6	21.1	80.0	17.8	80.0	27.9	62.6	28.2	75.0	33.8	34.2	↑0.9	↑3.1
	Token	699	1105	850	7588	896	6497	958	1908	1458	941	743	1215	+4.2%	-10.8%
REST (ours), COMPOSITION OF PROPERTIES															
Best pair	Acc.	75.5	87.0	27.8	83.3	21.1	86.7	29.5	60.1	28.2	85.0	32.3	41.7	↑3.2	↑7.5
	Token	794	1227	926	10281	981	9828	1039	2629	1330	1259	780	2134	+8.8%	+26.7%
All properties	Acc.	76.1	86.6	32.2	78.9	16.7	86.7	27.8	63.6	30.9	81.7	34.0	42.1	↑3.7	↑6.8
	Token	783	1253	941	11033	1023	11275	1086	2673	1377	1373	755	2151	+10.9%	+37.9%

D.4 Auxiliary-Loss Baseline Sweeps

Table 4 in the main text reports each auxiliary-loss baseline at its best available weight. Both CODI and SIM-CoT were run as full weight sweeps rather than a single point. We report the full grids here.

As shown in Table 12 and Table 13, neither baseline is sensitive to its auxiliary weight. Accuracy varies by less than a point across the full range of the CODI distillation weight and of the SIM-CoT step weight on the larger benchmarks, and the wider variation on the two AIME sets corresponds to one or two problems. The operating point reported in Table 4 is therefore representative of each method rather than an artifact of tuning.

D.5 Representation Analysis

Every analysis of this subsection and of Section 4.4 inspects a representative run of each arm rather than the full seed and weight grid of the tables above, since the quantities it reports describe the geometry of a trained thought rather than a benchmark score.

Diversity Ratio. We check whether $\mathbf{T}$'s own positions are diverse from one another. We measure diversity ratio

$$R(\mathbf{T}) = \frac{\text{Dist}(\mathbf{T})}{\sqrt{2}\text{DistVC}(\mathbf{T})}, \quad (72)$$

where

$$\text{Dist}(\mathbf{T}) = \frac{2}{m'(m'-1)} \sum_{i < j} \|\mathbf{T}_i - \mathbf{T}_j\|_2 \quad (73)$$

Table 11. Full property sweep, multi-agent (round $r = 1$), Light vs Scaled.

Method	Metric	Math500		AIME2025		AIME2026		GPQA-D		MedQA		Code Gen.		Avg. Change	
		Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled	Light	Scaled
CE only	Acc. Token	71.1 550	87.7 1018	22.2 849	76.7 9009	17.8 884	86.7 8488	24.9 749	63.1 2379	28.9 868	80.2 1137	32.5 604	39.0 1507	Base Base	Base Base
		REST (ours), CAUSALITY													
$\beta = 0.1$	Acc. Token	76.1 608	86.6 999	31.1 940	86.7 10437	20.0 1009	83.3 10161	26.6 941	65.2 2187	28.2 1151	83.3 1066	33.3 613	41.5 1882	$\uparrow 3.0$ +16.9%	$\uparrow 2.2$ +13.6%
$\beta = 0.3$	Acc. Token	77.8 601	86.4 1041	27.8 894	80.0 9538	22.2 978	85.6 9430	28.6 888	63.1 2285	30.0 1049	82.7 1115	33.9 643	39.5 1545	$\uparrow 3.8$ +12.2%	$\uparrow 0.7$ +6.0%
$\beta = 1.0$	Acc. Token	75.1 601	87.6 1034	27.8 880	83.3 10156	18.9 1063	86.7 9942	29.0 963	62.6 2089	30.1 1127	82.0 1119	32.6 685	42.1 1868	$\uparrow 2.7$ +18.1%	$\uparrow 1.8$ +11.3%
$\beta = 3.0$	Acc. Token	77.0 598	86.8 1022	27.8 915	78.9 9946	17.8 982	90.0 9436	26.9 955	64.1 2069	29.1 1106	84.0 1052	33.4 622	42.7 1968	$\uparrow 2.4$ +15.0%	$\uparrow 2.2$ +8.3%
$\beta = 0.1$	Acc. Token	REST (ours), MINIMALITY													
		74.1 610	87.2 1052	26.7 922	83.3 10181	16.7 990	90.0 9858	27.1 865	64.6 2292	27.9 1087	83.3 1079	34.1 626	42.6 1695	$\uparrow 1.5$ +13.3%	$\uparrow 2.9$ +11.1%
$\beta = 0.3$	Acc. Token	77.1 605	88.0 1044	30.0 920	83.3 10309	22.2 1052	90.0 9956	27.6 849	67.2 2287	26.8 1158	81.3 1120	32.0 575	42.3 2036	$\uparrow 3.1$ +14.6%	$\uparrow 3.1$ +13.7%
$\beta = 1.0$	Acc. Token	76.1 596	87.0 1072	32.2 902	86.7 10099	21.1 995	84.4 11072	28.5 871	65.7 1996	27.3 1093	83.7 1093	26.6 785	43.7 2075	$\uparrow 2.4$ +16.4%	$\uparrow 3.0$ +16.4%
$\beta = 3.0$	Acc. Token	75.3 593	87.2 1053	25.6 906	80.0 10527	20.0 1006	86.7 10111	25.1 867	60.6 2257	28.8 1162	83.3 1080	31.0 613	43.1 1955	$\uparrow 1.4$ +14.3%	$\uparrow 1.2$ +14.6%
$\beta = 0.1$	Acc. Token	REST (ours), SEPARABILITY													
		74.7 564	85.6 790	25.6 838	56.7 6648	17.8 923	60.0 6180	25.6 758	53.5 1084	30.9 1116	77.7 582	31.1 1650	30.0 744	$\uparrow 1.4$ +29.9%	$\downarrow 11.6$ -31.9%
$\beta = 0.3$	Acc. Token	73.6 548	84.8 728	24.4 881	63.3 5972	16.7 982	40.0 4337	26.9 825	56.1 971	27.7 1169	78.3 598	28.8 1756	29.3 795	$\uparrow 0.1$ +36.8%	$\downarrow 13.6$ -43.1%
$\beta = 1.0$	Acc. Token	71.3 549	80.2 718	26.7 948	60.0 4209	16.7 961	40.0 4255	24.2 807	58.6 893	26.3 1168	78.3 567	32.2 790	32.1 1034	0.0 +16.0%	$\downarrow 14.0$ -50.4%
$\beta = 3.0$	Acc. Token	72.3 570	84.6 824	26.7 930	70.0 5987	17.8 1014	70.0 6944	29.1 907	57.1 1316	29.4 1234	79.0 708	29.3 656	35.0 1307	$\uparrow 1.2$ +17.9%	$\downarrow 6.3$ -27.4%
$\beta = 0.1$	Acc. Token	REST (ours), STABILITY													
		70.1 495	85.8 983	22.2 773	73.3 7608	17.8 888	56.7 6961	28.3 729	61.6 1380	26.0 913	81.0 867	32.3 1219	31.8 797	$\downarrow 0.1$ +11.4%	$\downarrow 7.2$ -21.0%
$\beta = 0.3$	Acc. Token	74.7 586	83.4 806	28.9 904	53.3 6763	17.8 992	70.0 7196	27.4 929	59.6 1602	26.4 1247	79.3 509	28.7 1565	33.6 993	$\uparrow 1.1$ +38.2%	$\downarrow 9.0$ -24.1%
$\beta = 1.0$	Acc. Token	74.0 551	79.0 581	27.8 881	40.0 5220	16.7 976	46.7 5369	26.8 785	52.0 836	31.0 1205	77.3 445	35.1 781	31.6 882	$\uparrow 2.3$ +15.0%	$\downarrow 17.8$ -43.4%
$\beta = 3.0$	Acc. Token	72.1 546	79.8 586	24.4 884	46.7 4005	18.9 964	50.0 4327	24.2 825	53.0 701	29.8 1274	77.7 468	34.0 813	32.3 1092	$\uparrow 1.0$ +17.8%	$\downarrow 15.6$ -52.5%

Table 12. CODI $\mathcal{L}_{\text{KD}}$ weight sweep (round $r = 1$), Light.

β	Metric	Math500	AIME2025	AIME2026	GPQA-D	MedQA	Code Gen.
1	Acc.	76.2	30.0	15.6	26.4	28.9	32.7
	Token	598	9179	10298	931	1215	661
10	Acc.	76.9	26.7	17.8	24.2	28.9	32.9
	Token	596	9371	10308	877	1158	699
20	Acc.	76.9	30.0	22.2	28.8	27.3	32.0
	Token	598	9328	9867	861	1101	682

and

$$\text{DistVC}(\mathbf{T}) = \frac{1}{m'} \sum_{i=1}^{m'} \|\mathbf{T}_i - \mu\|_2 \quad (74)$$

are the mean pairwise distance and the mean distance to the consumer’s vocabulary centroid μ among $\mathbf{T}$ ’s m' positions. The $\sqrt{2}$ normalizer follows from concentration of measure, hence $R(\mathbf{T}) = 1$ matches a thought as diverse as a random sample of real vocabulary tokens and $R(\mathbf{T}) = 0$ is total collapse. Figure 8 shows Minimality, Causality, and their combinations have higher ratios than the

Table 13. SIM-CoT λ_{step} weight sweep (round $r = 1$), Light.

λ_{step}	Metric	Math500	AIME2025	AIME2026	GPQA-D	MedQA
0.3	Acc.	76.7	27.8	21.1	26.9	28.9
	Token	598	9287	10042	914	1202
1	Acc.	74.9	28.9	18.9	27.9	27.9
	Token	574	8696	9377	866	1055
3	Acc.	76.4	28.3	18.3	27.0	29.0
	Token	596	8945	10209	902	1159

CE-only baseline, Separability is close to it, and Stability has a lower ratio. This is consistent with Minimality and Causality’s losses spanning many positions of $\mathbf{T}$ while Separability and Stability reduce it to one pooled vector before applying any loss.

Effective Superposition. We measure how many candidate reasoning paths $\mathbf{T}$ supports at once, adapting the effective global parallelism of Deng et al. [13]. For a question, let $x^{(1)}, \dots, x^{(M)}$ denote M candidate refined plans of lengths $L_1, \dots, L_M$, namely the reference refined plan and the producer’s sampled plans that lead the consumer to the correct answer and remain distinct from one another. Let W denote the consumer’s unembedding matrix, and let $p_i = \text{softmax}(W\mathbf{T}_i)$ denote the token distribution decoded from position i of $\mathbf{T}$. The decoded mass that $\mathbf{T}$ places on a token x , pooled over its m' positions, is

$$\mu(x) = \sum_{i=1}^{m'} p_i[x] \mathbb{I}[x \in \mathcal{K}(p_i)], \quad (75)$$

where $\mathcal{K}(p_i)$ is the set of the K most probable tokens under p_i . Each candidate receives the coverage score

$$c_m = \frac{1}{L_m} \sum_{j=1}^{L_m} \mu(x_j^{(m)}), \quad (76)$$

which a tempered softmax converts into a posterior over the candidates,

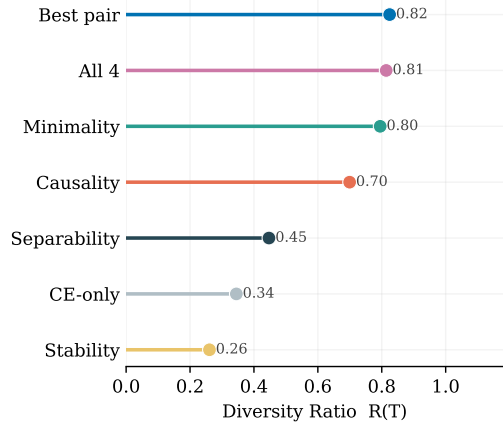
$$P_m = \frac{\exp(\log(c_m + \varepsilon)/\tau)}{\sum_{k=1}^M \exp(\log(c_k + \varepsilon)/\tau)}. \quad (77)$$

The effective superposition is the exponentiated entropy of that posterior,

$$N_{\text{eff}} = \exp\left(-\sum_{m=1}^M P_m \log P_m\right), \quad (78)$$

which equals one when a single candidate holds all of the mass and M when every candidate holds the same. We fix the same values for $K = 100$ and $\tau = 1$ as introduced in Deng et al. [13].

Deng et al. [13] align each latent step with a segment of every candidate. A freely generated $\mathbf{T}$ has no such correspondence. We therefore pool the decoded mass over all positions of $\mathbf{T}$ in eq. (75), normalize by the length of the candidate in eq. (76), and admit a top- K token at any position. A question on which every candidate has $c_m = 0$ yields a uniform posterior by construction, which would record an absence of evidence as maximal superposition, and such a question is excluded.


Figure 8. Vector diversity within $\mathbf{T}$.

Decoded Thought Content. Figure 9 decodes each transferred thought through the consumer’s vocabulary and compares the top-10 decoded tokens against the producer’s output. REST thoughts cover a wider set of distinct tokens, while CE-only concentrates on a small repeated set. The rate at which the decoded tokens match the producer’s output is also higher under REST. This is the behaviour eq. (2) targets.

Training CE. As shown in Figure 11(a), CE-only memorizes the training set, while causality retains a higher training CE and a higher accuracy. Stopping CE-only early, at the training CE where REST plateaus, does not recover the accuracy of REST in the Scaled system at $r = 1$. The gain of REST is therefore not explained by avoiding memorization. Figure 11(b) showcases the relation between training CE relative to CE-only and the change in accuracy. A lower training CE does not indicate a higher accuracy, which is the limitation of answer-level supervision in Section 2.1.

Questions Inside One Cluster. Figure 10 projects the thoughts of a fixed subsample of the training distribution, for which no held-out split exists. At the refiner to solver transfer on math, CE-only thoughts spread along a one-dimensional curve rather than a point, although most questions on it still share a near-identical thought with another question. We take the tightest cluster the CE-only objective produces at the planner to refiner transfer and inspect the questions behind it. The three below belong to that cluster, shortened to their statements, and each is labelled by its index in the training dataset.

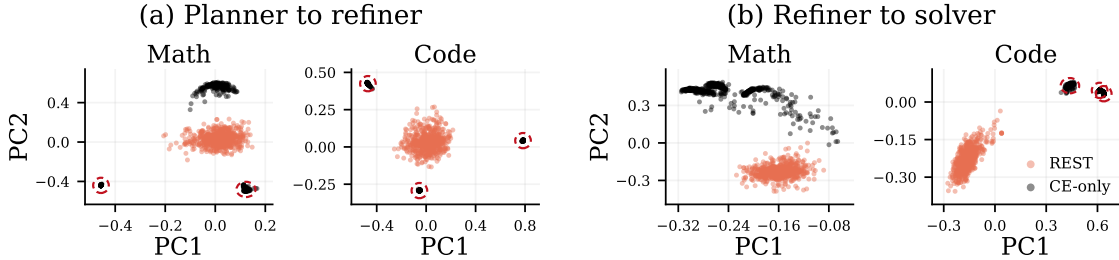


Figure 10. PCA for T. CE thoughts collapse into dense clusters, where each dashed ring holds all questions of one training run, while REST spreads thoughts apart.

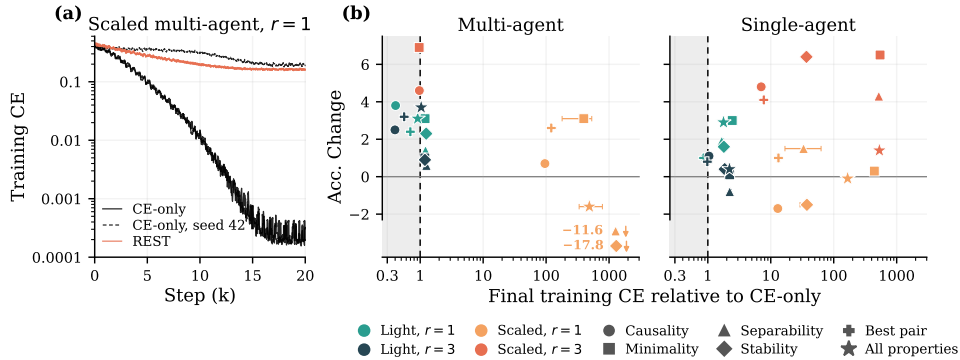


Figure 11. Training CE. (a) Training CE loss for causality at $\beta = 3.0$. (b) Mean of last CE 200 steps relative to CE-only against Acc. Change. Bars span seeds.

Their topics are a trigonometric series, refraction through a prism, and a bound state in quantum mechanics, and the cluster holds three further questions on combinatorial game theory, a recurrence

relation, and a construction over the real numbers. The questions do not share a domain nor a method of solution. Therefore, we conclude that the proximity of their thoughts represents collapse.

Under the CE-only objective the six thoughts of this cluster span 2.8% of the projection, measured as the largest distance between any two of them as a fraction of the largest distance between any two of the 200 thoughts. The same six questions span 56.4% under separability at $\beta = 3.0$. The questions grouped by the CE-only objective share no more content than unrelated ones. Under a six-cluster partition of the projection, taken across both transfers and all three seeds, the questions inside a cluster have a mean pairwise TF-IDF cosine similarity of 0.03, against 0.02 for groups of the same sizes drawn at random from the same pool.

Three Questions From One CE-only Cluster (Planner to Refiner Transfer, seed 42)

Sample 1832, trigonometric series.

Given that $A_k = \frac{k(k-1)}{2} \cos \frac{k(k-1)\pi}{2}$, find $|A_{19} + A_{20} + \dots + A_{98}|$.

Sample 967, refraction through a prism.

For an isosceles prism of angle A and refractive index μ , the angle of minimum deviation is $\delta_m = A$. Which of the following options is or are correct? [A] For the angle of incidence $i_1 = A$, the ray inside the prism is parallel to the base of the prism. (Three further options omitted.)

Sample 1553, bound state in quantum mechanics.

A particle of mass m moves in a one-dimensional potential $V(x) = -\alpha \delta(x)$, where $\delta(x)$ is the Dirac delta function and α is a positive constant. The particle is bound. Find the value of x_0 such that the probability of finding the particle with $|x| < x_0$ is exactly $1/2$.

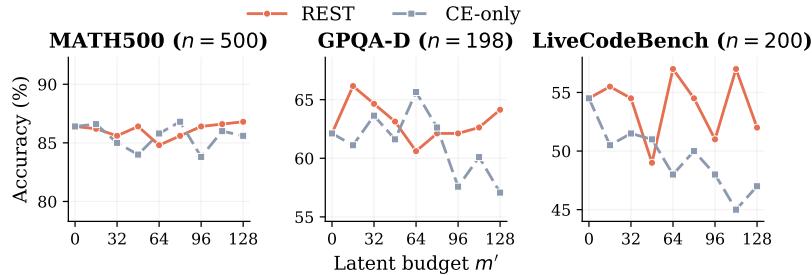


Figure 12. Accuracy against the latent budget m' for the Scaled system at $r = 1$.

Latent Budget. Figure 12 varies the latent budget m of the Scaled system at a fixed recursion round and compares REST against the CE-only objective. The two objectives stay close at the smallest budgets. As the budget grows, the CE-only objective loses accuracy on GPQA-D and on LiveCodeBench while REST holds its level, and neither objective separates on MATH500. Each cell is a single run per arm, and the variation between adjacent budgets is therefore wider than in the tables above.

E Case Study: Effect on Generation

To study the difference in tokens when using REST compared to CE, we first analyzed the single agent system. Since the single agent system contained shorter outputs on average, we looked at the samples that are shorter by at least 200 tokens. On those answers we measured how much of each answer repeats a token span. We sweep the span width from 4 to 24 tokens and the result stays the same. We report the midpoint percentage drops across span widths in Table 14, averaged over the four property terms, which agree on every benchmark. The tokens REST saves are repeated text.

Table 14. Single-agent repeated text.

Benchmark	Answers ≥ 200 tokens shorter	Repeated text	
		CE	REST
Math500	10–12%	4.5%	2.2%
AIME2025	21–23%	9.2%	3.4%
AIME2026	27–32%	8.7%	4.0%
GPQA-D	24–27%	28.1%	6.7%
MedQA	24–33%	51.5%	8.9%

The two boxes below follow one of these questions in full. The CE-only objective commits to polynomial long division and rewrites the full tableau at every elimination step, exhausting the token budget before reaching an answer. REST applies synthetic division and answers correctly in less than half the tokens. The shorter answer follows a different method (not a truncation of the other).

Output of the Single-Agent System, CE-only Objective (MATH500, sample 51, seed 42)

Question:

Find the quotient when $x^6 - 3$ is divided by $x + 1$.

Final Answer of the CE-only Objective:

To find the quotient when $x^6 - 3$ is divided by $x + 1$, we can use polynomial long division. Here are the steps:

1. **Set up the division:** $x + 1 \overline{) x^6 \ 0 \ 0 \ 0 \ 0 \ 0 \ -3}$

2. **Divide the leading term of the dividend by the leading term of the divisor:** $\frac{x^6}{x} = x^5$. Write x^5 above the division bar.

3. **Multiply x^5 by $x + 1$:** $x^5 \cdot (x + 1) = x^6 + x^5$. Write this product under the dividend and subtract:

$$\begin{array}{r} x+1 \overline{) x^6 \ 0 \ 0 \ 0 \ 0 \ 0 \ -3} \\ \underline{-(x^6 + x^5)} \\ 0 \ -x^5 \ 0 \ 0 \ 0 \ 0 \ -3 \end{array}$$

4. **Repeat the process with the new polynomial $-x^5$:** $\frac{-x^5}{x} = -x^4$. Write $-x^4$ above the division bar.

5. **Multiply $-x^4$ by $x + 1$:** $-x^4 \cdot (x + 1) = -x^5 - x^4$. Write this product under the current polynomial and subtract:

[steps 6 through 8 continue the same pattern, each restating the whole tableau]

9. **Multiply $-x^2$ by $x + 1$:** $-x^2 \cdot (x + 1) = -x^3 - x^2$. Write this product under the current polynomial and subtract: $\backslash[\ \backslash\begin{array}{r} \text{Budget exhausted, no quotient stated } \times$

Output of the Single-Agent System, REST (MATH500, sample 51, seed 42)

Question:

Find the quotient when $x^6 - 3$ is divided by $x + 1$.

Final Answer of REST:

To find the quotient when $x^6 - 3$ is divided by $x + 1$, we can use polynomial long division or synthetic division. Here, I will use synthetic division for efficiency.

1. **Set up the synthetic division:** The divisor is $x + 1$, so the root is -1 . The coefficients of the dividend $x^6 - 3$ are $1, 0, 0, 0, 0, 0, -3$ (we include 0 for the missing x^5, x^4, x^3 , and x^2 terms).

2. **Perform the synthetic division:** Bring down the first coefficient: 1. Multiply 1 by -1 and add to the next coefficient: $0 + (-1) = -1$. Multiply -1 by -1 and add to the next coefficient: $0 + 1 = 1$. Multiply 1 by -1 and add to the next coefficient: $0 + (-1) = -1$. Multiply -1 by -1 and add to the next coefficient: $0 + 1 = 1$. Multiply 1 by -1 and add to the next coefficient: $0 + (-1) = -1$. Multiply -1 by -1 and add to the next coefficient: $-3 + 1 = -2$.

The quotient is the polynomial formed by the numbers above the line, starting from the second coefficient: $x^5 - x^4 + x^3 - x^2 + x - 1$. The remainder is the last number: -2 .

Therefore, the quotient when $x^6 - 3$ is divided by $x + 1$ is $x^5 - x^4 + x^3 - x^2 + x - 1$ *Correct Answer ✓*

The multi-agent system shows the opposite pattern on some questions. There REST spends more tokens than CE-only, and the extra length is genuine derivation rather than repeated text. The boxes below follow one such question from GPQA-D. The answer options appear in the order the run assigns, since the benchmark options are permuted per run. The CE-only objective states an answer with no derivation. REST works through the relevant physics before answering, and only REST reaches the correct option. The derivation carries a slight arithmetic mistake at one intermediate step, and the answer is still correct.

Output of the Multi-Agent System, CE-only Objective (GPQA-D, sample 0, seed 42)

Question:

Two quantum states with energies E_1 and E_2 have a lifetime of 10^{-9} sec and 10^{-8} sec, respectively. We want to clearly distinguish these two energy levels. Which one of the following options could be their energy difference so that they can be clearly resolved?

A. 10^{-8} eV B. 10^{-11} eV C. 10^{-4} eV D. 10^{-9} eV

Final Answer of the CE-only Objective:

The correct answer is B. *Wrong Answer ✗*

Output of the Multi-Agent System, REST (GPQA-D, sample 0, seed 42)

Question:

Two quantum states with energies E_1 and E_2 have a lifetime of 10^{-9} sec and 10^{-8} sec, respectively. We want to clearly distinguish these two energy levels. Which one of the following options could be their energy difference so that they can be clearly resolved?

A. 10^{-8} eV B. 10^{-11} eV C. 10^{-4} eV D. 10^{-9} eV

Final Answer of REST:

To determine the energy difference between the two states that allows them to be clearly resolved, we need to use the relationship between energy and temperature in statistical physics, specifically the Fermi-golden rule and the linewidth of the energy levels.

The linewidth (width of the energy level) is given by the formula $\Delta E = \hbar \frac{\pi}{\Delta t}$, where $\hbar$ is the reduced Planck's constant and Δt is the lifetime of the level.

Given the lifetimes of the energy levels, $\Delta t_1 = 10^{-9}$ s and $\Delta t_2 = 10^{-8}$ s, the corresponding linewidths are $\Delta E_1 = \hbar\pi/\Delta t_1 \approx 3.30 \times 10^{-25}$ J and $\Delta E_2 = \hbar\pi/\Delta t_2 \approx 3.30 \times 10^{-24}$ J. Converting to eV gives $\Delta E_1 \approx 2.06 \times 10^{-6}$ eV and $\Delta E_2 \approx 2.06 \times 10^{-5}$ eV.

To clearly distinguish the two energy levels, the energy difference between the two states should be greater than the (larger) linewidth of the levels, so greater than roughly 2.06×10^{-5} eV. Comparing against the options, only 10^{-4} eV clears this bound.

The correct option is C *Correct Answer ✓*